

$$f(x) = \frac{7 \cos^4 x}{16 \sin x}$$

Now,

$$g(x) = \frac{1-x(1+|1-x|)}{|1-x|} f\left(\frac{1}{1-x} \frac{\sin\left(\frac{1}{1-x}\right)}{\cos^3\left(\frac{1}{1-x}\right)}\right)$$

$$g(x) = \frac{1-x(1+|1-x|)}{|1-x|} \cos\left(\frac{1}{1-x}\right)$$

$$\lim_{x \rightarrow 1^+} \frac{1-x(1-1+x)}{-1+x} \cos\left(\frac{1}{1-x}\right)$$

$$\lim_{x \rightarrow 1^+} \frac{1-x^2}{x-1} \cos\left(\frac{1}{1-x}\right)$$

$$\lim_{h \rightarrow 0} \frac{1-(1+h)^2}{1+h-1} \cos\left(\frac{1}{1-1-h}\right)$$

$$\lim_{h \rightarrow 0} \frac{-h \times (2+h)}{h} \cos\left(\frac{1}{h}\right)$$

$$\lim_{h \rightarrow 0} (h+2) \cos\left(\frac{1}{h}\right) \rightarrow \text{does not exist}$$

$$\lim_{x \rightarrow 1^-} g(x) \rightarrow \lim_{h \rightarrow 0} f(1-h)$$

$$\lim_{h \rightarrow 0} \frac{1-(1-h)(1+h)}{h} \cos\left(\frac{1}{h}\right)$$

$$\lim_{h \rightarrow 0} \underbrace{h \cos\left(\frac{1}{h}\right)}_{[-1,1]} \Rightarrow 0$$

D09/2/094. If $g: \mathbb{R} \rightarrow \mathbb{N} \cup \{0\}$ and $g(x) = n$ where 'x' represents the area of triangle joining the two fixed points $P(5, 0)$,

$Q(8, 4)$ and a variable point $R(p, q)$ such that $\angle PRQ = \frac{\pi}{2}$ and 'n' represents the numbers of such triangles.

Then which of the following is/are true :

(A) $g(5) = 4$

(B) $g(7) = 0$

(C) $g(6.25) = 2$

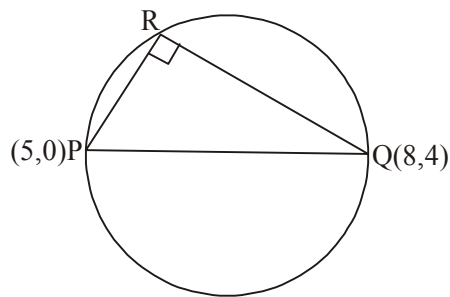
(D) $g(9) = 1$

Ans. ABC

Sol. Point R lies on a circle with PQ as a diameter.

$$\therefore PQ = 5 \Rightarrow r = \frac{5}{2}$$

$$\text{Maximum area of } \Delta PQR = \frac{1}{2} PQ \times r = r^2 = \frac{25}{4}$$



D09/2/095. Let $F(x)$ be a continuous function, satisfying $F(x) = \frac{1}{x^2} \int_{\sqrt{2}}^x \{4t^3 - 2F'(t)\} dt \quad \forall x \neq 0$ then :

- (A) The value of $F(0)$ is -2 .
 (B) $F(x)$ has exactly one maxima and no minima
 (C) The curve $y = F(x)$ is always concave upwards
 (D) Rolle's theorem is applicable to the function $g(x) = F(x) - F'(x)$ in $[0, 2]$

Ans. ACD

Sol. $x^2 F'(x) + 2x F(x) = 4x^3 - 2F'(x) \Rightarrow (x^2 + 2) F'(x) + 2x F(x) = 4x^3$

$$\Rightarrow \frac{d}{dx} \left\{ (x^2 + 2) F(x) \right\} = 4x^3 \quad \Rightarrow (x^2 + 2) F(x) = x^4 + c$$

From given relation, $f(\sqrt{2}) = 0 \therefore c = -4 \quad F(x) = \frac{x^4 - 4}{x^2 + 2} = x^2 - 2 \quad \therefore F(0) = -2$

$F(x) = x^2 - 2$ has exactly one minima at $x = 0$ and no maxima.

$F''(x) = 2 > 0 \therefore y = F(x)$ is always concave upward.

$$g(x) = F(x) - F'(x) = x^2 - 2x - 2$$

$$\therefore g(0) = g(2) = -2 \quad \therefore \text{Rolle's theorem is applicable in } [0, 2]$$

D09/2/096. $f(x) = (4\sin^2 x - 1)^n (x^2 + 6x + 11)$ where $n \in \mathbb{N}$, then $x = \frac{\pi}{6}$ is a point of :

- (A) local maximum if n is even
 (B) decreasing if n is odd
 (C) local minimum if n is even
 (D) increasing if n is odd

$f(x) = (4\sin^2 x - 1)^n (x^2 + 6x + 11)$ जहाँ $n \in \mathbb{N}$, तब $x = \frac{\pi}{6}$ एक बिन्दु है :

- (A) स्थानीय उच्चिष्ठ यदि n सम है
 (B) ह्रासमान यदि n विषम है
 (C) स्थानीय निम्निष्ठ यदि n सम है
 (D) वर्धमान यदि n विषम है

Ans. CD

Sol. $x^2 + 6x + 11 > 0 \quad \forall x \in \mathbb{R}$

$$f\left(\frac{\pi}{6}\right)=0, f\left(\frac{\pi^+}{6}\right)=\left(0^+\right)^n > 0, f\left(\frac{\pi^-}{6}\right)=\left(0^-\right)^n < 0$$

D09/2/097. A movie screen on a wall is 20 feet high and 10 feet above the floor. If a man has to position himself at distance x from the screen to have a maximum angle of view θ , then :

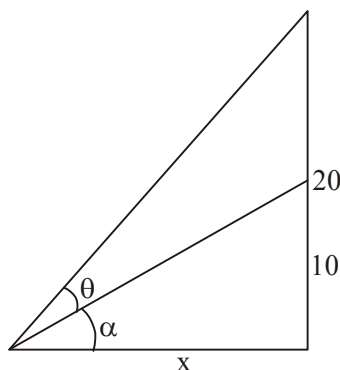
- (A) $x = \frac{10}{\sqrt{3}}$ feet (B) $\theta = 60^\circ$ (C) $x = 10\sqrt{3}$ feet (D) $\theta = 30^\circ$

एक फिल्म का पर्दा जिसकी ऊँचाई 20 मी. है, सतह से 10 मी. की ऊँचाई पर है। एक आदमी पर्दे से दूरी x पर खड़ा है ताकि पर्दे को देखने का कोण अधिकतम θ हो तब :

- (A) $x = \frac{10}{\sqrt{3}}$ फीट (B) $\theta = 60^\circ$ (C) $x = 10\sqrt{3}$ फीट (D) $\theta = 30^\circ$

Ans. CD

Sol.



$$\tan(\theta + \alpha) = \frac{30}{x} \quad \dots(i)$$

$$\tan \alpha = \frac{10}{x} \quad \dots(ii)$$

$$\frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha} = \frac{30}{x} \Rightarrow \tan \theta = \frac{20x}{x^2 + 300} = f(x)$$

$$\text{Now } \theta \rightarrow \max \Rightarrow \tan \theta \rightarrow \max \quad \{\because \theta < 90^\circ\}$$

$$f'(x) = 0 \Rightarrow x = 10\sqrt{3} \Rightarrow \theta = 30^\circ$$

D09/2/098. $f(x)$ is an even function, $x = 1$ is a point of minima and $x = 2$ is a point of maxima for $y = f(x)$. Further

$$\lim_{x \rightarrow \infty} f(x) = 0 \text{ and } \lim_{x \rightarrow 0} f(x) = \infty. f(x) \text{ is increasing in } [1, 2] \text{ \& decreasing everywhere in } (0, 1) \text{ \& } (2, \infty).$$

Also, $f(1) = 3$ & $f(2) = 5$. Further $f(x)$ is a differentiable function $\forall x \in \mathbb{R} - \{0\}$, then :

- (A) $f(x) = 0$ has no real roots
 (B) $y = f(x)$ and $y = |f(x)|$ are identical functions
 (C) $f'(x) = 0$ has exactly four real roots whose sum is zero
 (D) $f'(x) = 0$ has exactly four real roots whose sum is 6

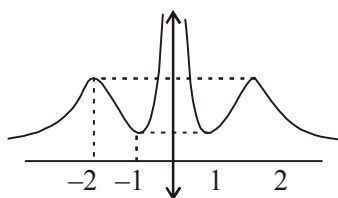
$f(x)$ एक सम फलन है, $x = 1$ फलन का निम्निष्ठ है तथा $x = 2$ फलन का उच्चिष्ठ है। $\lim_{x \rightarrow \infty} f(x) = 0$ तथा $\lim_{x \rightarrow 0} f(x) = \infty$.

$f(x)$ अंतराल $[1, 2]$ में वर्धमान है तथा अंतराल $(0, 1)$ & $(2, \infty)$ में ह्रासमान है। $f(1) = 3$ & $f(2) = 5$ है एवं $f(x)$ प्रत्येक $\forall x \in \mathbb{R} - \{0\}$ के लिए अवकलनीय है, तब :

- (A) $f(x) = 0$ का कोई वास्तविक मूल नहीं है
 (B) $y = f(x)$ तथा $y = |f(x)|$ तुल्य फलन है
 (C) $f'(x) = 0$ के चार वास्तविक मूल हैं, जिनका योग शून्य है
 (D) $f'(x) = 0$ के चार वास्तविक मूल हैं, जिनका योग 6 है

Ans. AB

Sol.



$$\lim_{x \rightarrow 0} f(x) = \infty, \quad \lim_{x \rightarrow \infty} f(x) = 0$$

$f(x)$ is increasing in $(1, 2)$

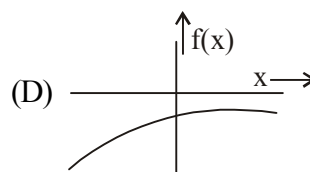
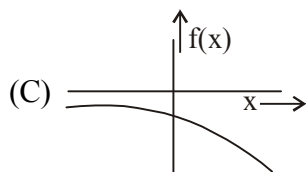
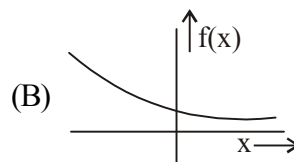
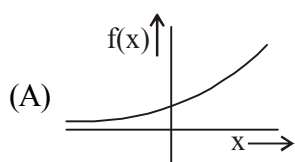
and decreasing in $(0, 1) \cup (2, \infty)$

from the graph

(C) is incorrect because $f'(x)$ could be zero at some other values also.

Hence (A), (B)

D09/2/099. Let $f(x) f'(x) f''(x) > 0 \forall x \in \mathbb{R}$, then which of the following is(are) best possible graph of $f(x)$:



Ans. AD

Sol. (A) $f(x) > 0, f'(x) > 0$ & $f''(x) > 0$

- (B) $f(x) > 0, f'(x) < 0$ & $f''(x) > 0$
 (C) $f(x) < 0, f'(x) < 0$ & $f''(x) < 0$
 (D) $f(x) < 0, f'(x) > 0$ & $f''(x) < 0$

D09/2/100. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 2x^2 \ln|x| - 5x^2, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Which of the following statement(s) is(are) correct?

- (A) $f(x)$ has exactly one local maximum and two local minimum points
 (B) $f(x)$ has strictly increasing $(10, \infty)$
 (C) Absolute minimum value of $f(x)$ exist but absolute maximum value of $f(x)$ does not exist
 (D) In the interval $k \in (0, \infty)$, $f(x) = k$ has two distinct roots

Ans. ABCD

Sol. $f(x)$ is clearly an even function.

For $x > 0$, $f(x) = x^2 \ln x^2 - 5x^2$

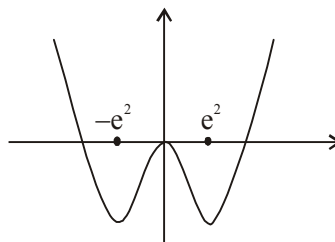
$f'(x) = 2x(\ln x^2 - 4)$

$f'(x) < 0$ for $x \in (0, e^2)$

$f'(x) > 0$ for $x \in (e^2, \infty)$

Using this info, we can sketch graph of $f(x)$.

Graphically, or otherwise, (ABC) are correct.



D09/2/101. Let $f(x) = a \sin x + b\sqrt{1-a^2} \cos x + c$, where $|A| < 1, b > 0$, then :

- (A) maximum value of $f(x)$ is b , if $c = 0$
 (B) difference of maximum and minimum value of $f(x)$ is $2b$
 (C) $f(x) = c$, if $x = -\cos^{-1}a$
 (D) $f(x) = c$, if $x = \cos^{-1}a$

Ans. ABC

Sol. $f(x) = \sqrt{a^2 b^2 + b^2 - b^2 a^2} \sin(x + \alpha) + c = b \sin(x + \alpha) + c$, where $\tan \alpha = \frac{\sqrt{1-a^2}}{a}$

$(f(x))_{\max} - (f(x))_{\min} = 2b$

Also, at $x = -\cos^{-1}a$, $f(x) = c$

D09/2/102. Let ABCD be a quadrilateral inscribed in a circle. Suppose $AB = \sqrt{2 + \sqrt{2}}$ and aB subtends 135° at the centre of the circle, the maximum possible area of ABCD is :

(A) $\frac{5+3\sqrt{3}}{4\sqrt{2}}$

(B) $\frac{8+\sqrt{7}}{\sqrt{3}}$

(C) $\frac{19+\sqrt{3}}{2}$

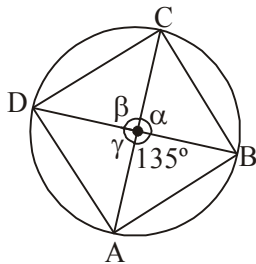
(D) none of these

Ans. A

Sol. $\because \alpha, \beta, \gamma < \pi$ and $y = \sin x$ is concave downward in $(0, \pi)$

$$\text{So, } \frac{\sin \alpha + \sin \beta + \sin \gamma}{3} \leq \sin \left(\frac{\alpha + \beta + \gamma}{3} \right)$$

$$\Rightarrow \frac{\sin \alpha + \sin \beta + \sin \gamma}{3} \leq \sin(75^\circ)$$



$$\text{So, area of } ABCD \leq \frac{1}{2} \sin(135^\circ) + \frac{1}{2} (\sin \alpha + \sin \beta + \sin \gamma)$$

$$= \frac{5+3\sqrt{3}}{4\sqrt{2}}$$

Equality holds if $\alpha = \beta = \gamma = 75^\circ$

D09/2/103. A factory D is to be connected by a road with a straight railway line on which a town A is situated. The distance DB of the factory to the railway line is $5\sqrt{3}$ km. Length AB of the railway line is 20 km. Freight charges on the road are twice the charges on the railway. The point P ($AP < AB$) on the railway line should the road DP be connected so as to ensure minimum freight charges from the factory to the town is :

(A) BP = 5km

(B) AP = 5km

(C) BP = 7.5 km

(D) None of these

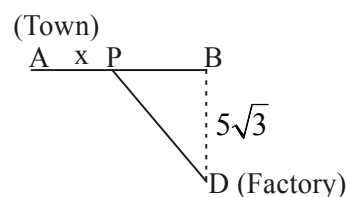
Ans. A

Sol. Let the charges for railway line be k Rs/km.

$$\text{Now, total freight charges, } T = kx + 2k\sqrt{(20-x)^2 + 75}$$

$$\text{Let } \frac{dT}{dx} = 0$$

$$\text{Or } k + 2k \frac{2(x-20)}{2\sqrt{(x-20)^2 + 75}} = 0$$



$$\begin{aligned}\text{Or } 4(x-20)^2 &= 75 + (x-20)^2 \\ \text{Or } (x-20)^2 &= 25 \\ \text{Or } x &= 25, 15 \text{ or } x = 15 \text{ (as } AP < AB) \\ \text{Or } PB &= AB - AP = 20 - 15 = 5\text{km}\end{aligned}$$

D09/2/104. Let $x = a$ be a point where $f(x) = 2|x| + |x+2| - |2|x| - |x+2||$ has a local extremum, and

$$\beta = \min \left\{ k \in \mathbb{R} \mid 4kx^2 + \frac{1}{x} \geq 1 \forall x > 0 \right\}.$$

Then which of the following option(s) is/are possible ?

- (A) $54\beta = -a$ (B) $81\beta = -a$ (C) $54\beta = -3a$ (D) $27\beta = -2a$

Ans. AC

Sol.

D09/2/105. Let $f: \mathbb{R} \rightarrow \mathbb{R}^+$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable functions such that f'' and g'' are continuous functions on $\mathbb{R}$. Suppose $(f'(2), g(2)) = (0, 0) \neq (f''(2), g'(2))$,

$$\lim_{x \rightarrow 2} \frac{f(x)g(x)}{f'(x)g'(x)} = 1 \text{ and } h(x) = \begin{vmatrix} \cos 2x & \cos 2x & \sin 2x \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix}.$$

Then which of the following options is/are INCORRECT ?

- (A) f has local minimum at $x = 2$, and h' has exactly three solutions in $(-\pi, \pi)$
 (B) f has local maximum at $x = 2$, and h attains maximum at $x = 0$
 (C) $f''(2) > f(2)$, and h attains minimum at $x = 0$
 (D) $f(x) = f''(x)$ has at least one real root, and h' has more than 3 roots in $(-\pi, \pi)$

Ans. AC

Sol.

D06/2/106. If $f(x) = \begin{vmatrix} x^n & \sin x & -\cos x \\ n! & \sin\left(\frac{n\pi}{2}\right) & \cos\left(\frac{n\pi}{2}\right) \\ a & a^2 & a^3 \end{vmatrix}$ the value of $\frac{d^n}{dx^n} (f(x))$ at $x = 0$ for $n = 2m + 1$ is :

- (A) -1 (B) 0 (C) a^6 (D) independent of a

यदि $f(x) = \begin{vmatrix} x^n & \sin x & -\cos x \\ n! & \sin\left(\frac{n\pi}{2}\right) & \cos\left(\frac{n\pi}{2}\right) \\ a & a^2 & a^3 \end{vmatrix}$, $x = 0$ पर $n = 2m + 1$ के लिए $\frac{d^n}{dx^n}(f(x))$ का मान होगा :

(A) -1

(B) 0

(C) a^6

(D) a से स्वतंत्र

Ans.

BD

Sol.

$$f(x) = \begin{vmatrix} x^n & \sin x & -\cos x \\ x! & \sin\left(\frac{n\pi}{2}\right) & \cos\left(\frac{n\pi}{2}\right) \\ a & a^2 & a^3 \end{vmatrix}$$

Case-I

$n = 2m + 1$ and $m = \text{odd}$

$\Rightarrow n \in \{3, 7, 11, \dots\}$

$$f(x) = \begin{vmatrix} x^{2m+1} & \sin x & -\cos x \\ (2m+1)! & -1 & 0 \\ a & a^2 & a^3 \end{vmatrix}$$

$$\frac{d^n f(x)}{dx^n} = \begin{vmatrix} (2m+1)! & -\cos x & -\sin x \\ (2m+1)! & -1 & 0 \\ a & a^2 & a^3 \end{vmatrix}$$

$$\left. \frac{d^n f(x)}{dx^n} \right|_{x=0} = \begin{vmatrix} (2m+1)! & -1 & 0 \\ (2m+1)! & -1 & 0 \\ a & a^2 & a^3 \end{vmatrix} = 0$$

Case-2

$n = 2m + 1$ and $m = \text{even}$

$n \in \{5, 9, 13, \dots\}$

$$f(x) = \begin{vmatrix} x^{2m+1} & \sin x & -\cos x \\ (2m+1)! & 1 & 0 \\ a & a^2 & a^3 \end{vmatrix}$$

$$\frac{d^n f(x)}{dx^n} = \begin{vmatrix} (2m+1)! & \cos x & \sin x \\ (2m+1)! & 1 & 0 \\ a & a^2 & a^3 \end{vmatrix} \Rightarrow \left. \frac{d^n f(x)}{dx^n} \right|_{x=0} = 0$$

D06/2/107. $f_n(x) = e^{f_{n-1}(x)}$ for all $n \in \mathbb{N}$ and $f_0(x) = x$, then $\frac{d}{dx}\{f_n(x)\}$ is :

- (A) $f_n(x) \frac{d}{dx}\{f_{n-1}(x)\}$ (B) $f_n(x) f_{n-1}(x)$
 (C) $f_n(x) f_{n-1}(x) \dots f_2(x) \cdot f_1(x)$ (D) None of these

$f_n(x) = e^{f_{n-1}(x)} \forall n \in \mathbb{N}$ तथा $f_0(x) = x$ है, तब $\frac{d}{dx}\{f_n(x)\}$ होगा :

- (A) $f_n(x) \frac{d}{dx}\{f_{n-1}(x)\}$ (B) $f_n(x) f_{n-1}(x)$
 (C) $f_n(x) f_{n-1}(x) \dots f_2(x) \cdot f_1(x)$ (D) इनमें से कोई नहीं

Ans. AC

Sol. $\frac{d}{dx}\{f_n(x)\} = \frac{d}{dx}\{e^{f_{n-1}(x)}\}$
 $= e^{f_{n-1}(x)} \frac{d}{dx}\{f_{n-1}(x)\}$
 $= f_n(x) \cdot \frac{d}{dx}\{e^{f_{n-2}(x)}\} = f_n(x) \cdot e^{f_{n-2}(x)} \frac{d}{dx}\{f_{n-2}(x)\}$
 $= f_n(x) f_{n-1}(x) \frac{d}{dx}\{f_{n-2}(x)\}$
 $= f_n(x) f_{n-1}(x) \dots f_2(x) \frac{d}{dx}\{f_1(x)\}$
 $= f_n(x) f_{n-1}(x) \dots f_2(x) e^{f_0(x)} \frac{d}{dx}\{f_0(x)\}$
 Use $e^{f_0(x)} = f_1(x)$ and $f_0(x) = x$

D06/2/108. If $f(x) = x^3 + 3x + 4$ and g is the inverse function of $f(x)$, then the value of $\frac{d}{dx}\left(\frac{g(x)}{g(g(x))}\right)$ at $x = 4$ equals :

- (A) $\frac{-1}{6}$ (B) 6 (C) $\frac{-1}{3}$ (D) non-existent

यदि $f(x) = x^3 + 3x + 4$ तथा $g, f(x)$ का व्युत्क्रम फलन है, तब $x = 4$ पर $\frac{d}{dx}\left(\frac{g(x)}{g(g(x))}\right)$ का मान होगा :

- (A) $\frac{-1}{6}$ (B) 6 (C) $\frac{-1}{3}$ (D) विद्यमान नहीं

Ans. C

Sol.
$$\frac{d}{dx} \left(\frac{g(x)}{g(g(x))} \right) = \frac{g(g(x)) \cdot g'(x) - g(x) \cdot g'(g(x)) \cdot g'(x)}{(g(g(x)))^2}$$

Now, $f(0) = 4$

$$\Rightarrow f^{-1}(4) = g(4) = 0 \quad \text{And} \quad g'(4) = \frac{1}{f'(0)} = \frac{1}{3}$$

$$\text{Also, } f(-1) = 0 \quad \Rightarrow f^{-1}(0) = g(0) = -1$$

$$\begin{aligned} \therefore \frac{d}{dx} \left(\frac{g(x)}{g(g(x))} \right)_{x=4} &= \frac{g(g(4))g'(4) - g(4)g'(g(4))g'(4)}{(g(4))^2} \\ &= \frac{g(0) \cdot \frac{1}{f'(0)} - 0}{(g(0))^2} \end{aligned}$$

$$= \frac{-1 \times \frac{1}{3}}{1}$$

$$= \frac{-1}{3}$$

D06/2/109. Let $f_n(\alpha)$ denote the n^{th} derivative of function f at $x = \alpha$. Also

$$f(x) = -\frac{x^3}{3} + x^2 \sin \frac{3a}{2} - x \sin a \sin 2a - 5 \sin^{-1}(a^2 - 8a + 17),$$

$$g(x) = h\left(x - \sqrt{1-x^2}\right), h'(x) = 1 - x^2 \text{ and } p(x) = \frac{3x-1}{x^2-1}.$$

Then, which of the following options is/are correct ?

माना $f_n(\alpha)$, $x = \alpha$ पर f का $n^{\text{वाँ}}$ अवकलज दर्शाता है।

$$f(x) = -\frac{x^3}{3} + x^2 \sin \frac{3a}{2} - x \sin a \sin 2a - 5 \sin^{-1}(a^2 - 8a + 17),$$

$$g(x) = h\left(x - \sqrt{1-x^2}\right), h'(x) = 1 - x^2 \text{ तथा } p(x) = \frac{3x-1}{x^2-1}.$$

तब, निम्न में से कौनसा/से विकल्प सही है/हैं ?

$$(A) f'(\sin 8) < 0 \quad (B) 25g'\left(\frac{3}{5}\right) = 42 \quad (C) p_{50}(0) = 50! \quad (D) f'(\sin 8) > 0$$

Ans. ABC

Sol.
$$f(x) = -\frac{x^3}{3} + x^2 \sin \frac{3a}{2} - x \sin a \sin 2a - 5 \sin^{-1}(a^2 - 8a + 17)$$

$$-1 \leq a^2 - 8a + 17 \leq 1$$

$$-1 \leq (a-4)^2 + 1 \leq 1$$

$$\Rightarrow a = 4$$

$$f'(x) = -x^2 + 2x \sin \frac{3a}{2} - \sin a \sin 2a$$

$$= -\left(x - \sin \frac{3a}{2}\right)^2 + \sin^2 \frac{3a}{2} - \sin a \sin 2a$$

$$= -\left(x - \sin \frac{3a}{2}\right)^2 + \left(\frac{1 - \cos 3a}{2}\right) - (\cos a - \cos 3a)$$

$$f'(x) = -\left(x - \sin \frac{3a}{2}\right)^2 + \sin^2 \frac{a}{2}$$

$$f'(\sin 8) = -[\sin 8 - \sin 12]^2 + \sin^2 2$$

$$= -[2 \sin 2 \cos 10]^2 + \sin^2 2$$

$$f'(\sin 8) = (\sin^2 2) [1 - 2\cos^2 10] < 0$$

$$g(x) = h\left(x - \sqrt{1-x^2}\right)$$

$$g'(x) = h'\left(x - \sqrt{1-x^2}\right) \times \left[1 + \frac{x}{\sqrt{1-x^2}}\right]$$

$$g'(x) = \left[1 - \left(x - \sqrt{1-x^2}\right)^2\right] \left[\frac{x + \sqrt{1-x^2}}{\sqrt{1-x^2}}\right]$$

$$= \left[1 - \left(x^2 + 1 - x^2 - 2x\sqrt{1-x^2}\right)\right] \left[\frac{x + \sqrt{1-x^2}}{\sqrt{1-x^2}}\right]$$

$$= \left[2x\sqrt{1-x^2} \right] \left[\frac{x + \sqrt{1-x^2}}{\sqrt{1-x^2}} \right]$$

$$g'(x) = 2x \left(x + \sqrt{1-x^2} \right)$$

$$g'\left(\frac{3}{5}\right) = 2 \times \frac{3}{5} \left(\frac{3}{5} + \frac{4}{5} \right)$$

$$25g'\left(\frac{3}{5}\right) = 42$$

$$P(x) = \frac{3x-1}{x^2-1}$$

$$= (1-3x)(1-x^2)^{-1}$$

$$P_{50}(x) = 50!$$

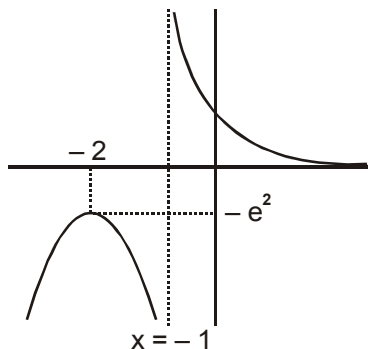
D08/2/110. For the equation $\frac{e^{-x}}{1+x} = \lambda$ which of the following statement(s) is/are **CORRECT** ?

- (A) When $\lambda \in (0, \infty)$ equation has 2 real and distinct roots
- (B) When $\lambda \in (-\infty, -e^2)$ equation has 2 real and distinct roots
- (C) When $\lambda \in (0, \infty)$ equation has 1 real root.
- (D) When $\lambda \in (-e, 0)$ equation has no real root.

Ans. BCD

Sol. $f'(x) > 0 \quad \frac{(1+x)e^{-x} - e^{-x}}{(1+x)^2} > 0$

for $x < -2$ Increasing.



D08/2/111. If $|f''(x)| \leq 1 \quad \forall x \in \mathbb{R}$, and $f(0) = 0 = f'(0)$, then which of the following can not be true :

यदि $|f''(x)| \leq 1 \quad \forall x \in \mathbb{R}$, तथा $f(0) = 0 = f'(0)$, तो निम्न में से कौनसा सत्य नहीं हो सकता :

$$(A) f\left(-\frac{1}{2}\right) = \frac{1}{6}$$

$$(B) f(2) = -4$$

$$(C) f(-2) = 3$$

$$(D) f\left(\frac{1}{2}\right) = \frac{1}{5}$$

Ans. ABCD

Sol. $-1 \leq f'(x) \leq 1$

$$-x \leq f(x) \leq x \quad (\because f(0) = 0)$$

$$-\frac{x^2}{2} \leq f(x) \leq \frac{x^2}{2} \quad (\because f(0) = 0)$$

D08/2/112. Let $f(x) = 3 \sin x - 4 \cos x + ax + b$, then :

(A) $f(x) = 0$ has only one real root which is positive if $a > 5$ and $b < 0$

(B) $f(x) = 0$ has only one real root which is negative if $a > 5$ and $b > 0$

(C) $f(x) = 0$ has only one real root which is negative if $a < -5$ and $b < 0$

(D) $f(x) = 0$ has only one real root which is positive if $a < -5$ and $b > 0$

माना $f(x) = 3 \sin x - 4 \cos x + ax + b$, तब :

(A) $f(x) = 0$ का केवल एक वास्तविक मूल है जो धनात्मक है यदि $a > 5$ और $b < 0$

(B) $f(x) = 0$ का केवल एक वास्तविक मूल है जो ऋणात्मक है यदि $a > 5$ और $b > 0$

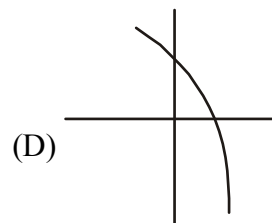
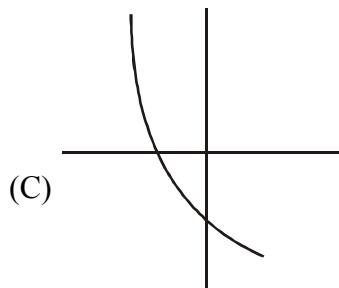
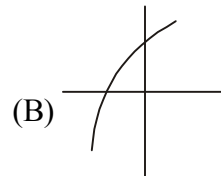
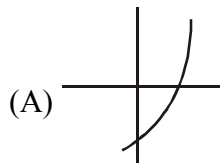
(C) $f(x) = 0$ का केवल एक वास्तविक मूल है जो ऋणात्मक है यदि $a < -5$ और $b < 0$

(D) $f(x) = 0$ का केवल एक वास्तविक मूल है जो धनात्मक है यदि $a < -5$ और $b > 0$

Ans. AC

Sol. $f'(x) = 3 \cos x + 4 \sin x + a > 0$ if $a > 5$

$$f'(x) < 0 \text{ if } a < -5$$



D08/2/113. If $f(x) = \frac{x}{1+x} - \ln(1+x)$ for $x > 0$ then :

(A) $f(x)$ is increasing function

(B) $f(x)$ is decreasing function

(C) $\frac{\pi - e}{(1+e)(1+\pi)} > \ln\left(\frac{1+\pi}{1+e}\right)$

(D) $\frac{\sin 1 - \cos 1}{(1+\sin 1)(1+\cos 1)} < \ln\left(\frac{1+\sin 1}{1+\cos 1}\right)$

यदि $f(x) = \frac{x}{1+x} - \ln(1+x)$, $x > 0$ के लिए तब :

(A) $f(x)$ वर्धमान फलन है

(B) $f(x)$ ह्यासमान फलन है

(C) $\frac{\pi - e}{(1+e)(1+\pi)} > \ln\left(\frac{1+\pi}{1+e}\right)$

(D) $\frac{\sin 1 - \cos 1}{(1+\sin 1)(1+\cos 1)} < \ln\left(\frac{1+\sin 1}{1+\cos 1}\right)$

Ans. BD

Sol. (A) Let $f(x) = \frac{x}{1+x} - \ln(1+x)$ $x \in (0, \infty)$

$$f'(x) = -\frac{x}{(1+x)^2} < 0 \quad \forall x > 0$$

hence $f(x)$ is decreasing function

for C and D option apply $f(e) > f(\pi)$ & $f(\sin 1) < f(\cos 1)$

माना $f(x) = \frac{x}{1+x} - \ln(1+x)$ $x \in (0, \infty)$

$$f'(x) = -\frac{x}{(1+x)^2} < 0 \quad \forall x > 0$$

अतः $f(x)$ ह्यासमान फलन है

विकल्प C और D के लिए $f(e) > f(\pi)$ & $f(\sin 1) < f(\cos 1)$

D08/2/114. Let $f(x)$ be a differentiable function with $f(1)f(-1) \neq 0$. Define a function $g(x) = \frac{x^2 - 1}{f(x)}$. If $g(x)$ does not

follow Rolle's theorem in $[-1, 1]$, then which of the following options is/are false ?

(A) $f(x) = 0$ cannot have any root in $[-1, 1]$

(B) $f(x) = 0$ has at least one root in $[-1, 1]$

(C) $f'(x)$ is zero at atleast one point in $[-1, 1]$

(D) $f(x)$ cannot satisfy Rolle's theorem in $[-1, 1]$

माना कि $f(x)$ अवकलनीय फलन है तथा $f(1)f(-1) \neq 0$ और फलन $g(x) = \frac{x^2 - 1}{f(x)}$ परिभाषित है। यदि $g(x)$, $[-1, 1]$ में रोल

प्रमेय को लागू नहीं करता है। तब निम्न में से कौनसा विकल्प गलत है/हैं ?

(A) $f(x) = 0$ का $[-1, 1]$ में कोई मूल नहीं हो सकता है

(B) $f(x) = 0$ का कम से कम एक मूल $[-1, 1]$ में है

(C) $[-1, 1]$ में कम से कम एक बिन्दु पर $f'(x)$ शून्य है

(D) $f(x)$ अंतराल $[-1, 1]$ में रोल प्रमेय को संतुष्ट नहीं करता है

Ans. ACD

Sol. $g(x)$ is not necessarily be defined in the complete interval $[-1, 1]$ and $g(x)$ will satisfy Rolle's theorem unless $f(x) = 0$ for some $x \in (-1, 1)$.

So far as $f'(x)$ is concerned, without further information nothing definite can be said.

Hindi : फलन $g(x)$, $[-1, 1]$ के अंतराल में निश्चित रूप से पूर्णतया परिभाषित नहीं है। जब तक किसी भी $x \in (-1, 1)$ के लिए $f(x) \neq 0$ है, $g(x)$ रोल प्रमेय को संतुष्ट करेगा। जहाँ तक $f'(x)$ की बात है, किसी अन्य जानकारी के बिना इसके बारे में कुछ भी निश्चित रूप से नहीं कह सकते।

D08/2/115. Let a function 'f' is continuous and differentiable with $f(0) = 1$ and $f(1) = 2$. Then, for $\forall n \in \mathbb{N}, n \geq 2$, there exists atleast one $c \in [0, 1]$ such that :

माना फलन 'f' सतत् और अवकलनीय फलन है जबकि $f(0) = 1$ और $f(1) = 2$ तब सभी $\forall n \in \mathbb{N}, n \geq 2$, के लिए कम से कम एक $c \in [0, 1]$ इस प्रकार है कि :

- (A) $f'(c)[f(c)]^{n-1} > \sqrt{2^{n-1}}$ (B) $f'(c)[f(c)]^{n-1} < \sqrt{2^{n-1}}$
(C) $f'(c)[f(c)]^{n-1} > 1$ (D) $f'(c)[f(c)]^{n-1} < 1$

Ans. AC

Sol. Let $g(x) = [f(x)]^n$

As $g(x)$ is continuous and differentiable in $[0, 1]$

$\therefore$ By LMVT, there exists a $c \in [0, 1]$, such that $g'(c) = \frac{f(1) - f(0)}{1 - 0}$

$$\Rightarrow n[f(x)]^{n-1} \cdot f'(x) \Big|_{x=c} = \frac{2^n - 1}{1}$$

$$\Rightarrow nf'(c) \cdot [f(c)]^{n-1} = 1 + 2 + 2^2 + \dots + 2^{n-1}$$

$$\Rightarrow f'(c)[f(c)]^{n-1} = \frac{1 + 2 + 2^2 + \dots + 2^{n-1}}{n}$$

As $AM \geq GM$,

$$\therefore \frac{1 + 2 + 2^2 + \dots + 2^{n-1}}{n} > \sqrt[n]{2^{1+2+\dots+n-1}}$$

$$\Rightarrow \frac{1 + 2 + 2^2 + \dots + 2^{n-1}}{n} > 2^{\frac{(n-1)(n)}{2} \times \frac{1}{n}}$$

$$\therefore f'(c)[f(c)]^{n-1} > 2^{\frac{n-1}{2}}$$

Hindi : माना $g(x) = [f(x)]^n$

चूँकि $g(x)$ अंतराल $[0, 1]$ में सतत् और अवकलनीय है

∴ By LMVT, there exists a $c \in [0, 1]$, such that $g'(c) = \frac{f(1) - f(0)}{1 - 0}$

$$\Rightarrow n[f(x)]^{n-1} \cdot f'(x) \Big|_{x=c} = \frac{2^n - 1}{1}$$

$$\Rightarrow nf'(c) \cdot [f(c)]^{n-1} = 1 + 2 + 2^2 + \dots + 2^{n-1}$$

$$\Rightarrow f'(c)[f(c)]^{n-1} = \frac{1 + 2 + 2^2 + \dots + 2^{n-1}}{n}$$

As $AM \geq GM$,

$$\therefore \frac{1 + 2 + 2^2 + \dots + 2^{n-1}}{n} > \sqrt[n]{2^{1+2+\dots+n-1}}$$

$$\Rightarrow \frac{1 + 2 + 2^2 + \dots + 2^{n-1}}{n} > 2^{\frac{(n-1)(n)}{2} \times \frac{1}{n}}$$

$$\therefore f'(c)[f(c)]^{n-1} > 2^{\frac{n-1}{2}}$$

D08/2/116. Let $f(x) = \frac{x^3}{9-x^2}$ and ℓ = number of integers in the domain of $f(x)$, where $f(x)$ is increasing.

m = number of solutions of $f(x) = \sin x$

n = least positive integral value of k for which $f(x) = k$ posses exactly 3 different solutions,

then :

माना $f(x) = \frac{x^3}{9-x^2}$ तथा $\ell = f(x)$ के प्रांत में पूर्णाकों की संख्या

$m = f(x)$ के हलों की संख्या $= \sin x$

$n = k$ के न्यूनतम धनात्मक पूर्णाक मान जिसके लिए $f(x) = k$ के ठीक 3 भिन्न-भिन्न हल हैं।

$$(A) m^2 = \ell \quad (B) 5m = n^2 - \ell^2 \quad (C) n + \ell + m = 20 \quad (D) \ell n + m = 75$$

Ans. ACD

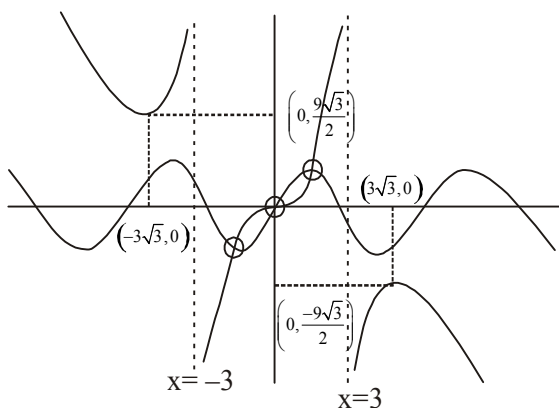
Sol. $f(x) = \frac{x^3}{9-x^2} \Rightarrow f'(x) = \frac{3x^2(9-x^2) - x^3(-2x)}{(9-x^2)^2} \Rightarrow f'(x) = \frac{-x^2(x^2-27)}{(x-3)^2(x+3)^2}$

For increasing $f(x)$ के वर्धमान के लिए, $f'(x) \geq 0$

$$f'(x) \geq 0 \Rightarrow x \in (-3\sqrt{3}, -3) \cup (-3, 3) \cup (3, 3\sqrt{3})$$

integers in that interval पूर्णांक अंतराल में स्थित हैं : $-5, -4, -2, -1, 0, 1, 2, 4, 5$

$$\Rightarrow \ell = 9$$



3 point of intersection प्रतिच्छेदन के तीन बिन्दु $\Rightarrow f(x) = \sin x$ have 3 roots के 3 मूल हैं

$$\Rightarrow m = 3$$

$f(x) = k$ has 3 solution for के 3 हल के लिए $k > \frac{9\sqrt{3}}{2}$

$$\Rightarrow n = 8$$

D08/2/117. If $f(x) = 3x^2 + \int_0^x e^{-t} f(x-t) dt$, then :

(A) $f(x) = 0$ has 3 solutions

(B) $f(x)$ is monotonic increasing

(C) $f(x) = 4$ has 2 solutions

(D) $\int_{-1}^1 f(x) dx = 2$

Ans. CD

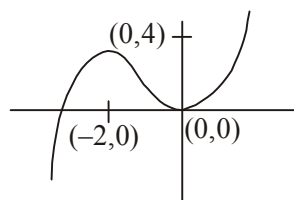
Sol.

$$f(x) = 3x^2 + e^{-x} \int_0^x e^t f(t) dt$$

$$f'(x) = 6x - e^{-x} \left(\frac{f(x) - 3x^2}{e^{-x}} \right) + e^{-x} e^x f(x)$$

$$f'(x) = 3x^2 + 6x$$

$$f(x) = x^3 + 3x^2 \quad [\because f(0) = 0]$$



D08/2/118. If f is an odd continuous in $[-1, 1]$ and differentiable in $(-1, 1)$, then which of the following statement(s) is(are) correct ?

- (A) $f'(a) = f(1)$ for some $a \in (-1, 0)$
 (B) $f'(b) = f(1)$ for some $b \in (0, 1)$
 (C) $n(f(\alpha))^{n-1} f'(\alpha) = f(f(1))^n$ for some $\alpha \in (-1, 0)$ and $\forall n \in \mathbb{N}$
 (D) $n(f(\beta))^{n-1} f'(\beta) = f(f(1))^n$ for some $\beta \in (0, 1)$ and $n \in \mathbb{N}$

Ans. ABD

Sol. Let $k = f(1)$

$$\Rightarrow f(-1) = -k$$

$$\text{and } f(0) = 0$$

(A) and (B) are both true using LMVT applied to $f(x)$ in $(-1, 0)$ and $(0, 1)$ respectively.

Also, consider $(f(x))^n = g(x)$

$$g(1) = k^n \text{ and } g(-1) = (-k)^n \Rightarrow g(-1) = \begin{cases} -k^n & n \text{ odd} \\ k^n & n \text{ even} \end{cases}$$

(D) is correct by applying LMVT to $(0, 1)$

(C) is incorrect for even n .

$$\text{E.g.: Let } f(x) = -x \text{ and } n = 2 \Rightarrow g(x) = x^2$$

$$\text{LHS: } g'(\beta) = n(f(\beta))^{n-1} f'(\beta) = 1(-\beta)(-1) = \beta$$

$$\text{RHS: } (f(1))^n = (-1)^2 = 1$$

$\beta \neq 1$ because $\beta \in (-1, 0)$

D08/2/119. Let $f(x) = \cot x - \tan x - 2 \tan 2x - 4 \tan 4x - 8 \cot 8x$, $x \neq \frac{n\pi}{8}$, $n \in \mathbb{I}$ and $g(x) = x^3 + 6x - 1$. One more

function $h(x)$ is defined as $h : \mathbb{R} - \left\{ \frac{n\pi}{8}, n \in \mathbb{I} \right\} \rightarrow \mathbb{R}$, $h(x) = f(x) + g(x)$ then identify the correct statement(s)

:

$$(A) \quad h''\left(\frac{\pi}{24}\right) = \frac{\pi}{4}$$

(B) $h(x)$ is odd function

(C) $h(x)$ is increasing in the domain

(D) If the equation $h(x) = \lambda$ has a solution in $(0, 3)$ then number of integral values of λ is 7

Ans. AC

Sol. $f(x) = \cot x - \tan x - 2 \tan 2x - 4 \tan 4x - 8 \tan 8x = 0 \quad \because \tan x + 2 \cot 2x = \cot x$

$$h(x) = f(x) + g(x)$$

$$h(x) = x^3 + 6x - 1, \quad x \in \mathbb{R} - \left\{ \frac{n\pi}{8}, n \in \mathbb{I} \right\}$$

$$(A) \quad h'(x) = 3x^2 + 6; \quad h''(x) = 6x; \quad h''\left(\frac{\pi}{24}\right) = \frac{\pi}{4}$$

(B) $h(x) = x^3 + 6x - 1$ which is not an odd function

(C) $h'(x) = 3x^2 + 6 > 0 \Rightarrow h(x)$ is increasing

(D) $h(x) = \lambda$, values of λ in $(0, 3)$ for which equation has a solution if $\lambda \in (-1, 43)$

D08/2/120. Given that f is a real valued non-constant differentiable function such that $f(x) \cdot f'(x) \leq 0$, for all real x , then it follows that :

(A) $f^2(x)$ is increasing function

(B) $f^2(x)$ is decreasing function

(C) $f(x)$ has no critical point

(D) $f(x) = 0$ does not have any real root

Ans. BCD

Sol. Case possible $f(x) > 0$ and $f(x)$ decreasing

or $f(x) < 0$, $f(x)$ increasing

and $f(x)$ and $f'(x)$ have to change sign simultaneously to maintain the condition which is not possible also $(f^2(x))' = 2f(x) \cdot f'(x) \leq 0$

D08/2/121. In a question a student was given to find the derivative of the product of two functions f and g . The student by mistake thought $(fg)' = f'g'$ for his question $f(x) = x^3$ and he got the correct answer. Given that $g(4) = 1$. Which of the following is/are correct possibilities (where $g(x) > 0$) :

(A) $g(x)$ is decreasing in $(-\infty, 3)$

(B) $g(x)$ is increasing in $(3, +\infty)$

(C) $g(x)$ is decreasing in $(3, +\infty)$

(D) $g(x)$ is increasing in $(-\infty, 3)$

Ans. CD

Sol. $g'(x) = \frac{-3}{x-3} g(x)$

$\Rightarrow \ln g(x) = -3 \ln|(x-3)| + c, c = 0$ as $g(4) = 1$

$$g(x) = \begin{cases} \frac{1}{(x-3)^3} & x > 3 \\ -\frac{1}{(x-3)^3} & x < 3 \end{cases}$$

D08/2/122. Let $f(x)$ be a non-negative continuous and bounded function for all $x \geq 0$. If $(\cos x) f'(x) \leq (\sin x - \cos x) f(x) \forall x \geq 0$, then which of the following is/are correct ?

(A) $f(6) + f(5) > 0$

(B) $x^2 - 3x + 2 + f(7) = 0$ has 2 distinct solutions

(C) $f(5)f(7) - f(6)f(5) = 0$

(D) $\lim_{x \rightarrow 6} \frac{f(x) - \sin(\pi x)}{(x-6)} = 1$

Ans. BC

Sol. Let $G(x) = e^x \cos x f(x)$
 $\therefore G'(x) = e^x \cos x f'(x) + e^x(\cos x - \sin x) f(x)$
 $= e^x (\cos x f'(x) + (\cos x - \sin x) f(x)) \leq 0$ (as given that $\cos x \cdot f'(x) \leq (\sin x - \cos x) f(x)$)
 $\therefore G(x)$ is a non increasing function.
 $\therefore G(6) = e^6 \cos 6 f(6) \leq 0$
 $\therefore f(6) \leq 0$
 But given that $f(x)$ is non-negative
 $\therefore f(6) = 0$
 With similar reasons $f(5), f(7) = 0$
 Thus $x^2 - 3x + 2 + f(7) = 0$
 Or $x^2 - 3x + 2 = 0$ has 2 distinct solutions

$$\lim_{x \rightarrow 6} \frac{f(x) - \sin(\pi x)}{(x - 6)}$$

$$= \lim_{x \rightarrow 6} \frac{f'(x) - \pi \cos(\pi x)}{1} \quad (\text{applying L'Hospital Rule})$$

$$= f'(6) - \pi < 0 \quad (\text{as } f'(6) \leq 0)$$

D08/2/123. Let $|X|$ denotes the cardinal number of set X ,

$$A = \{x \in \mathbb{R} \mid x^4 - 4x^3 + 12x^2 + x - 1 = 0\},$$

$$B = \{x \in \mathbb{R} \mid x^5 - 5x + a = 0\}.$$

Then, which of the following options is/are correct ?

- (A) $a > 2|A| \Rightarrow |B| = 1$ (B) $|B| = 1 \Rightarrow |a| > 2|A|$
 (C) $|a| < 2|A| \Rightarrow |B| = 3$ (D) $|B| = 3 \Rightarrow |a| \leq 2|A|$

Ans. ACD

Sol.

D08/2/124. For every twice differentiable function $f: \mathbb{R} \rightarrow [-2, 2]$ with $(f(0))^2 + (f'(0))^2 = 10$, which of the following options is/are correct ?

- (A) There exists interval $I_1 \subseteq \mathbb{R}$, and $I_2 \subseteq \mathbb{R}$ such that f is one-one in I_1 and many-one in I_2 .
 (B) There exists $\alpha \in (-4, 0)$ and $\beta \in (0, 4)$ such that $|f'(\alpha)| \leq 1$ and $|f'(\beta)| \leq 1$.
 (C) f must be a many-one function satisfying $\lim_{n \rightarrow \infty} f(n) = 1$.
 (D) There exists $\alpha \in (-4, 4)$ such that $f(\alpha) + f''(\alpha) = 0$ and $f'(\alpha) \neq 0$

Ans. ABD

Sol.

D07/2/125. Let $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ be a equation of curve, then which of the following is correct ?

(A) Equation of tangent at $P(x_1, y_1)$ to the curve is $\frac{x}{x_1^{\frac{1}{3}}} + \frac{y}{y_1^{\frac{1}{3}}} = a^{\frac{2}{3}}$

(B) Length of portion of tangent intercepted between coordinate axes is constant

(C) Equation of normal at $P(x_1, y_1)$ to the curve is $xx_1^{\frac{1}{3}} + yy_1^{\frac{1}{3}} = x_1^{\frac{4}{3}} + y_1^{\frac{4}{3}}$

(D) If tangent at (x_1, y_1) meet the coordinate axes in A and B, then locus of mid point of AB is $x^2 + y^2 = \frac{a^2}{4}$

माना वक्र का समीकरण $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ है तब निम्न में से कौनसा सही है ?

(A) वक्र के $P(x_1, y_1)$ पर स्पर्श रेखा का समीकरण $\frac{x}{x_1^{\frac{1}{3}}} + \frac{y}{y_1^{\frac{1}{3}}} = a^{\frac{2}{3}}$ है

(B) निर्देशांक अक्षों के मध्य स्पर्श रेखा के भाग की लम्बाई अचर है

(C) वक्र के बिन्दु $P(x_1, y_1)$ पर अभिलम्ब का समीकरण $xx_1^{\frac{1}{3}} + yy_1^{\frac{1}{3}} = x_1^{\frac{4}{3}} + y_1^{\frac{4}{3}}$ है

(D) यदि (x_1, y_1) पर स्पर्श रेखा, निर्देशांक अक्षों को A तथा B पर मिलती है तब AB के मध्य बिन्दु का बिन्दुपथ है $x^2 + y^2 = \frac{a^2}{4}$

Ans. ABD

Sol. Given curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

$$\text{slope of tangent} = \left. \frac{dy}{dx} \right|_{(x_1, y_1)} = -\frac{y_1^{\frac{1}{3}}}{x_1^{\frac{1}{3}}}$$

$$\text{slope of normal} = \frac{x_1^{\frac{1}{3}}}{y_1^{\frac{1}{3}}}$$

hence equation of tangent $y - y_1 = -\frac{y_1^{\frac{1}{3}}}{x_1^{\frac{1}{3}}}(x - x_1) \Rightarrow \frac{x}{x_1^{\frac{1}{3}}} + \frac{y}{y_1^{\frac{1}{3}}} = a^{\frac{2}{3}}$

now equation of normal $y - y_1 = \frac{x_1^{\frac{1}{3}}}{y_1^{\frac{1}{3}}}(x - x_1) \Rightarrow xx_1^{\frac{1}{3}} - yy_1^{\frac{1}{3}} = x_1^{\frac{4}{3}} - y_1^{\frac{4}{3}}$

now tangent meet x-axis at $A\left(a^{\frac{2}{3}}x_1^{\frac{1}{3}}, 0\right)$ and y-axis at $B\left(0, a^{\frac{2}{3}}y_1^{\frac{1}{3}}\right)$ and let mid-point of AB is M(h, k)

hence $h = \frac{a^{\frac{2}{3}}x_1^{\frac{1}{3}}}{2}$, $k = \frac{a^{\frac{2}{3}}y_1^{\frac{1}{3}}}{2}$

$x_1 = \left(\frac{2h}{a^{\frac{2}{3}}}\right)^3$, $y_1 = \left(\frac{2k}{a^{\frac{2}{3}}}\right)^3$

$x_1 = \frac{8h^3}{a^2}$, $y_1 = \frac{8k^3}{a^2}$

Now (x_1, y_1) satisfy given curve $x_1^{\frac{2}{3}} + y_1^{\frac{2}{3}} = a^{\frac{2}{3}}$

$\left(\frac{8h^3}{a^2}\right)^{\frac{2}{3}} + \left(\frac{8k^3}{a^2}\right)^{\frac{2}{3}} = a^{\frac{2}{3}}$

$\frac{4h^2}{a^{\frac{4}{3}}} + \frac{4k^2}{a^{\frac{4}{3}}} = a^{\frac{2}{3}}$

hence locus is $x^2 + y^2 = \frac{a^2}{4}$

Hindi. दिया गया वक्र $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

सपर्श रेखा की प्रवणता $= \frac{dy}{dx}\bigg|_{(x_1, y_1)} = -\frac{y_1^{\frac{1}{3}}}{x_1^{\frac{1}{3}}}$

$$\text{अभिलम्ब की प्रवणता} = \frac{x_1^{\frac{1}{3}}}{\frac{1}{y_1^3}}$$

$$\text{अतः स्पर्श रेखा की समीकरण } y - y_1 = -\frac{y_1^{\frac{1}{3}}}{x_1^{\frac{1}{3}}}(x - x_1) \Rightarrow \frac{x}{x_1^{\frac{1}{3}}} + \frac{y}{y_1^{\frac{1}{3}}} = a^{\frac{2}{3}}$$

$$\text{अभिलम्ब का समीकरण } y - y_1 = \frac{x_1^{\frac{1}{3}}}{y_1^{\frac{1}{3}}}(x - x_1) \Rightarrow xx_1^{\frac{1}{3}} - yy_1^{\frac{1}{3}} = x_1^{\frac{4}{3}} - y_1^{\frac{4}{3}}$$

$$\text{अब स्पर्श रेखा x-अक्ष को } A\left(a^{\frac{2}{3}}x_1^{\frac{1}{3}}, 0\right) \text{ पर तथा y-अक्ष को } B\left(0, a^{\frac{2}{3}}y_1^{\frac{1}{3}}\right) \text{ पर मिलती है। माना AB का मध्य बिन्दु } M(h, k)$$

$$\text{अतः } h = \frac{a^{\frac{2}{3}}x_1^{\frac{1}{3}}}{2}, k = \frac{a^{\frac{2}{3}}y_1^{\frac{1}{3}}}{2}$$

$$x_1 = \left(\frac{2h}{a^{\frac{2}{3}}}\right)^3, y_1 = \left(\frac{2k}{a^{\frac{2}{3}}}\right)^3$$

$$x_1 = \frac{8h^3}{a^2}, y_1 = \frac{8k^3}{a^2}$$

$$\text{अब } (x_1, y_1) \text{ वक्र } x_1^{\frac{2}{3}} + y_1^{\frac{2}{3}} = a^{\frac{2}{3}} \text{ को संतुष्ट करता है।}$$

$$\left(\frac{8h^3}{a^2}\right)^{\frac{2}{3}} + \left(\frac{8k^3}{a^2}\right)^{\frac{2}{3}} = a^{\frac{2}{3}}$$

$$\frac{4h^2}{a^{\frac{4}{3}}} + \frac{4k^2}{a^{\frac{4}{3}}} = a^{\frac{2}{3}}$$

$$\text{अतः बिन्दुपथ } x^2 + y^2 = \frac{a^2}{4} \text{ है।}$$

D07/2/126. Possible value(s) of 'a' for which curves C_1 ; $y = \sqrt{3} + \sin x$, $x \in (0, 2\pi)$ and C_2 ; $y = \frac{|x|}{2} + a$, have a common tangent is/are :

'a' के सम्भावित मान, जिसके लिए वक्रों C_1 ; $y = \sqrt{3} + \sin x$, $x \in (0, 2\pi)$ तथा C_2 ; $y = \frac{|x|}{2} + a$ की एक उभयनिष्ठ स्पर्शरेखा है, है/हैं :

- (A) $\frac{3\sqrt{3}}{2} - \frac{\pi}{6}$ (B) $\frac{3\sqrt{3}}{2} + \frac{\pi}{6}$ (C) $\frac{\sqrt{3}}{2} + \frac{5\pi}{6}$ (D) $\frac{\sqrt{3}}{2} - \frac{5\pi}{6}$

Ans. AD

Sol. Common tangent $\Rightarrow \frac{dy}{dx}$ same $\Rightarrow \cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}$ or $\frac{5\pi}{3}$ in $(0, 2\pi)$ at $x = \frac{\pi}{3}$

$$\Rightarrow y = \sqrt{3} + \frac{\sqrt{3}}{2} = \frac{\pi}{6} + a \Rightarrow a = \frac{3\sqrt{3}}{2} - \frac{\pi}{6}$$

$$\text{at } x = \frac{5\pi}{3} \Rightarrow y = \sqrt{3} - \frac{\sqrt{3}}{2} = \frac{5\pi}{6} + a \Rightarrow a = \frac{\sqrt{3}}{2} - \frac{5\pi}{6}.$$

Hence (A), (D)

D07/2/127. Tangent to the curve $x^n + y^n = a^n$ ($a > 0$) at $P(x, y)$ meets x-axis at Q and y-axis at R, ($x, y \neq 0$), then :

(A) If QR is constant for all P on the curve $\Rightarrow n = 1$ or $\frac{2}{3}$

(B) If OR + OQ is constant for all P on the curve $\Rightarrow n = 1$ or $\frac{1}{2}$

(C) If OR + OQ is constant for all P on the curve $\Rightarrow n = 1$ or $\frac{1}{3}$

(D) If QR is constant for all P on the curve $\Rightarrow n = 1$ or $\frac{1}{3}$

वक्र $x^n + y^n = a^n$ ($a > 0$) के बिन्दु $P(x, y)$ पर स्पर्श रेखा, x-अक्ष को Q पर तथा y-अक्ष को R, ($x, y \neq 0$) पर मिलती है तब :

(A) यदि QR वक्र पर सभी P के लिए अचर है $\Rightarrow n = 1$ या $\frac{2}{3}$

(B) यदि OR + OQ वक्र पर सभी P के लिए अचर है $\Rightarrow n = 1$ या $\frac{1}{2}$

(C) यदि OR + OQ वक्र पर सभी P के लिए अचर है $\Rightarrow n = 1$ या $\frac{1}{3}$

(D) यदि QR वक्र पर सभी P के लिए अचर है $\Rightarrow n = 1$ या $\frac{1}{3}$

Ans. AB

Sol. $xx_1^{n-1} + yy_1^{n-1} = a^n$ is tangent at P.

point Q is $\left(\frac{a^n}{x_1^{n-1}}, 0\right)$ and point R is $\left(0, \frac{a^n}{y_1^{n-1}}\right)$

$$OR + OQ = a^n(x_1^{1-n} + y_1^{1-n}) = \text{const} \Rightarrow 1 - n = 0 \text{ or } n \Rightarrow n = 1 \text{ or } \frac{1}{2}$$

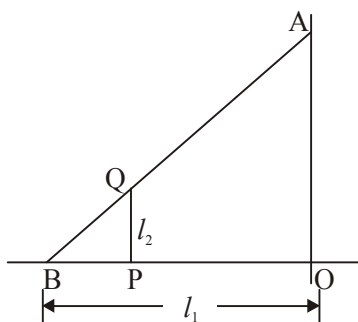
$$QR = a^n \sqrt{x_1^{2-2n} + y_1^{2-2n}} = \text{const} \Rightarrow 2 - 2n = 0 \text{ or } n \Rightarrow n = 1 \text{ or } \frac{2}{3}$$

D07/2/128. A lamp of negligible height is placed on the ground ℓ_1 m away from a wall. A man ℓ_2 m tall is walking at a speed of $\frac{\ell_1}{10}$ m/s from the lamp to the nearest point on the wall. When he is midway between the lamp and the wall, the rate of change in the in the length of this shadow on the wall is :

(A) $-\frac{5\ell_2}{2}$ m/s (B) $-\frac{2\ell_2}{5}$ m/s (C) $-\frac{\ell_2}{2}$ m/s (D) $-\frac{\ell_2}{5}$ m/s

Ans. B

Sol.



Let BP = x. From similar triangle property, we get

$$\frac{AO}{\ell_1} = \frac{\ell_2}{x} \text{ or } AO = \frac{\ell_1 \ell_2}{x} \text{ or } \frac{d(AO)}{dt} = \frac{-\ell_1 \ell_2}{x^2} \frac{dx}{dt}$$

$$\text{When } x = \frac{\ell_1}{2}, \frac{d(AO)}{dt} = -\frac{2\ell_2}{5} \text{ m/s.}$$

D07/2/129. Which of the following options is/are correct ?

(A) 1 is the length of x-intercept of tangent to both branches of function

$$f(x) = \begin{cases} -x^2, & x < 0 \\ x^2 + 8, & x \geq 0 \end{cases}$$

(B) If the area of triangle formed by tangent to $xy^n = a^{n+1}$ and coordinate axes is a constant, then $n = 1$

(C) The segment of normal to curve $x = 2a \sin t + a \sin t \cos^2 t$; $y = -a \cos^3 t$ contained between coordinate-axes is equal to $2a$.

(D) The area of triangle formed by tangent to $xy^n = a^{n+1}$ and coordinate axes is constant, for infinitely many real values of n .

Ans. ABC

Sol.

D07/2/130. Consider the following curves, where t is a parameter

$$C : x = \sqrt{a} \sec t, y = \sqrt{b} \csc t, D : x = \sec^2 t, y = \cot t.$$

If λ is the x-intercept of the tangent of C at any arbitrary point $P(\alpha, \beta)$, and tangent to D at $A\left(t = \frac{\pi}{4}\right)$ meets it again at B , then :

(A) λ is proportional to α^3

(B) $2AB = 3\sqrt{5}$

(C) λ is proportional to $\sqrt[3]{\alpha}$

(D) $3AB = 2\sqrt{5}$

Ans. AB

Sol.

D07/2/131. Let tangent to C at $P(1, 7)$ touches the circle D at (α, β) , where

$$C : y = x^2 + 6, D : x^2 + y^2 + 16x + 12y + c = 0, E : y = e^x.$$

If $A(c, e^c)$, $B(c-1, e^{c-1})$, $C(c+1, e^{c+1})$, then which of the following options is/are correct ?

(A) Tangent of E at A intersects AB on LHS of A

(B) $2\alpha - 3\beta = 9$

(C) Tangent of E at A intersects AB on RHS of A

(D) $3\alpha - 2\beta = 4$

Ans. AB

Sol.

COMPREHENSION TYPE QUESTIONS

Paragraph D05/3/132 to D05/3/134

Let $f(x) = f_1(x) - 2f_2(x)$ where $f_1(x) = \begin{cases} \min.(x^2, |x|), & -1 \leq x \leq 1 \\ x^2, & |x| > 1 \end{cases}$ and

$f_2(x) = \begin{cases} |x|, & |x| > 1 \\ \max.(x^2, |x|), & |x| \leq 1 \end{cases}$. Also $g(x) = \begin{cases} \min.\{f(t) : -3 \leq t \leq x\}, & \text{for } -3 \leq x < 0 \\ \max.\{f(t) : 0 \leq t \leq x\}, & \text{for } 0 \leq x < 3 \end{cases}$.

माना $f(x) = f_1(x) - 2f_2(x)$ जहाँ $f_1(x) = \begin{cases} \min.(x^2, |x|), & -1 \leq x \leq 1 \\ x^2, & |x| > 1 \end{cases}$ तथा

$f_2(x) = \begin{cases} |x|, & |x| > 1 \\ \max.(x^2, |x|), & |x| \leq 1 \end{cases}$ हैं। $g(x) = \begin{cases} \min.\{f(t) : -3 \leq t \leq x\}, & \text{for } -3 \leq x < 0 \\ \max.\{f(t) : 0 \leq t \leq x\}, & \text{for } 0 \leq x < 3 \end{cases}$.

D05/3/132. Range of $f(x)$, is :

$f(x)$ का परिसर है :

- (A) $(-\infty, -1]$ (B) $(-\infty, 1]$ (C) $[1, \infty)$ (D) $[-1, \infty)$

Ans. D

Sol.

D05/3/133. Number of values of x where $g(x)$ fails to be differentiable, is :

x के मानों की संख्या जहाँ $g(x)$ अवकलनीय नहीं है :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol.

D05/3/134. $\lim_{x \rightarrow 0} f(g(x))$ is equal to :

- (A) 0 (B) 1 (C) -1 (D) non existent

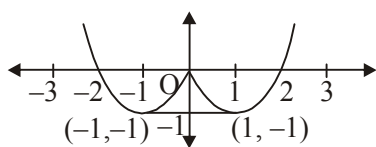
$\lim_{x \rightarrow 0} f(g(x))$ बराबर है :

- (A) 0 (B) 1 (C) -1 (D) विद्यमान नहीं

Ans. C

Sol. $f_1(x) = x \quad \forall x \in \mathbb{R}$

and $f_2(x) = |x| \quad \forall x \in \mathbb{R}$



Graph of $y = f(x)$

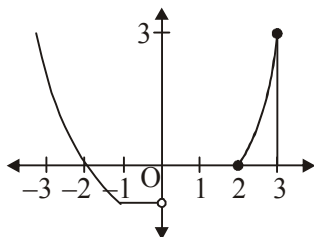
Now, $f(x) = f_1(x) - 2f_2(x)$

$\Rightarrow f(x) = x^2 - 2|x|$

$$\therefore f(x) = \begin{cases} x^2 - 2x, & x \geq 0 \\ x^2 + 2x, & x \leq 0 \end{cases}$$

(i) Clearly from above graph, range of $f(x) = [-1, \infty)$

$$\text{Now, } g(x) = \begin{cases} x^2 + 2x, & x \in [-3, -1) \\ -1, & x \in [-1, 0) \\ 0, & x \in [0, 2] \\ x^2 - 2x, & x \in (2, 3) \end{cases}$$



Graph of $y = g(x)$

(ii) $g(x)$ is non-derivable at $x = 0, 2$

(iii) For $\lim_{x \rightarrow 0} f(g(x))$, we have

$$\text{R.H.L.} = \lim_{h \rightarrow 0} f(g(0+h)) = \lim_{h \rightarrow 0} f(0) = 0$$

$$\text{L.H.L.} = \lim_{h \rightarrow 0} f(g(0-h)) = \lim_{h \rightarrow 0} f(-1) = -1$$

$\Rightarrow$ Limit does not exist.

Paragraph D05/3/135 to D05/3/136

Let f be a differentiable function on $\mathbb{R}$ satisfying $\frac{f(x+y) - f(x)}{2} = \frac{f(y) - 2}{2} + xy$, $\forall x, y \in \mathbb{R}$, and $f'(0) = 3$.

माना $f, \mathbb{R}$ पर एक अवकलनीय फलन है जो $\frac{f(x+y)-f(x)}{2} = \frac{f(y)-2}{2} + xy, \forall x, y \in \mathbb{R}$ को संतुष्ट करता है, तथा $f'(0) = 3$ है।

D05/3/135. The minimum value of $f(|x|)$ is equal to :

$f(|x|)$ का न्यूनतम मान होगा :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol.

D05/3/136. The number of points where $|f(|x|)|$ is non-derivable, is :

बिन्दुओं की संख्या जहाँ $|f(|x|)|$ अन-अवकलनीय है, है :

- (A) 0 (B) 1 (C) 2 (D) 5

Ans. B

Sol. (i) Given, $\frac{f(x+y)-f(x)}{2} = \frac{f(y)-2}{2} + xy \forall x, y \in \mathbb{R}$... (1)

$$\text{Now, } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} = \lim_{h \rightarrow 0} \left(\frac{f(h)-2+2 \times h}{h} \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{f(h)-2}{h} \right) + 2x$$

$$\Rightarrow f'(x) = f'(0) + 2x$$

(putting $x = 0, y = 0$ in equation (1), we get $f(0) = 2$)

$$\therefore f'(x) = 3 + 2x$$

Now, integrating both sides with respect to x , we get

$$\int f'(x) dx = \int (2 + 2x) dx \Rightarrow f(x) = 3x + x^2 + c$$

$$\text{As } f(0) = 2 \Rightarrow 2 = c \Rightarrow f(x) = f(x) = x^2 + 3x + 2$$

$$f(|x|) = |x|^2 + 3|x| + 2$$

$\Rightarrow$ The minimum value of $f(|x|)$ is equal to 2

$$(ii) \text{ As } y = f(|x|) = |x|^2 + 3|x| + 2 = \begin{cases} x^2 - 3x + 2, & x \leq 0 \\ x^2 + 3x + 2, & x > 0 \end{cases}$$

Also, $|f(|x|)| = f(|x|)$ and $y'(0^+) = 3, y'(0^-) = -3$

$\Rightarrow |f(|x|)|$ is non-derivable at one point viz. $x = 0$

Paragraph D05/3/137 to D05/3/138

Suppose f, g and h be three real valued function defined on $\mathbb{R}$.

$$\text{Let } f(x) = 2x + |x|, \quad g(x) = \frac{1}{3}(2x - |x|) \text{ and } h(x) = f(g(x))$$

माना f, g तथा h तीन $\mathbb{R}$ पर परिभाषित मान फलन है।

$$\text{माना } f(x) = 2x + |x|, \quad g(x) = \frac{1}{3}(2x - |x|) \text{ तथा } h(x) = f(g(x))$$

D05/3/137. The range of the function $k(x) = 1 + \frac{1}{\pi} (\cos^{-1}(h(x)) + \cot^{-1}(h(x)))$ is equal to :

फलन $k(x) = 1 + \frac{1}{\pi} (\cos^{-1}(h(x)) + \cot^{-1}(h(x)))$ का परिसर होगा :

- (A) $\left[\frac{1}{4}, \frac{7}{4}\right]$ (B) $\left[\frac{5}{4}, \frac{11}{4}\right]$ (C) $\left[\frac{1}{4}, \frac{5}{4}\right]$ (D) $\left[\frac{7}{4}, \frac{11}{4}\right]$

Ans. B

Sol.

D05/3/138. The function $T(x) = f(g(f(x))) + g(f(g(x)))$, is :

- (A) continuous and differentiable in $(-\infty, \infty)$
(B) continuous but not derivable $\forall x \in \mathbb{R}$
(C) neither continuous nor derivable $\forall x \in \mathbb{R}$
(D) an odd function

फलन $T(x) = f(g(f(x))) + g(f(g(x)))$:

- (A) $(-\infty, \infty)$ में सतत तथा अवकलनीय है
(B) $x \in \mathbb{R}$ के लिए सतत परन्तु अवकलनीय नहीं है
(C) $x \in \mathbb{R}$ के लिए न सतत न अवकलनीय है
(D) एक विषम फलन है

Ans. B

Sol. We have $f(x) = \begin{cases} 3x, & x \geq 0 \\ x, & x < 0 \end{cases}$ and $g(x) = \begin{cases} \frac{x}{3}, & x \geq 0 \\ x, & x < 0 \end{cases}$

Clearly f and g are inverse of each other

$$\text{Now, } h(x) = f(g(x)) = \begin{cases} 3\left(\frac{x}{3}\right) = x, & x \geq 0 \\ x, & x < 0 \end{cases}$$

(i) As $h(x) = x \quad \forall x \in \mathbb{R}$

$$\Rightarrow k(x) = 1 + \frac{1}{\pi} (\cos^{-1}x + \cot^{-1}x)$$

Domain of $k(x) = [-1, 1]$ and $k(x)$ is decreasing function on $[-1, 1]$

As $k(x)$ is continuous function on $[-1, 1]$

$$\text{Now, } k_{\min.}(x=1) = 1 + \frac{1}{\pi} (\cos^{-1}1 + \cot^{-1}1) = 1 + \frac{1}{\pi} \left(0 + \frac{\pi}{4}\right) = 1 + \frac{1}{4} = \frac{5}{4}$$

$$k_{\max.}(x=-1) = 1 + \frac{1}{\pi} (\cos^{-1}(-1) + \cot^{-1}(-1)) = 1 + \frac{1}{\pi} \left[\pi + \frac{3\pi}{4}\right] = 1 + \frac{7}{4} = \frac{11}{4}$$

$$\Rightarrow \text{Range of } k(x) = \left[\frac{5}{4}, \frac{11}{4}\right]$$

(iii) As f & g are inverse of each other,

$$\text{so } T(x) = f(g(f(x))) + g(f(g(x))) = f(x) + g(x) = (2x + |x|) + \frac{1}{3} (2x - |x|)$$

$$\Rightarrow T(x) = \begin{cases} \frac{10x}{3}, & x \geq 0 \\ 2x, & x < 0 \end{cases}$$

Clearly, $T(x)$ is continuous but non-derivable at $x = 0$

Paragraph D05/3/139 to D05/3/141

Consider, $f(x) = \frac{(x^2 - 1)(x^2 - 4)}{g(x)}$, $x \in \mathbb{R} - \{a, b\}$, where $g(x)$ is a polynomial of degree ≤ 4 . $f(x)$ has a

removable type of discontinuity at $x = a$ and $x = b$. Also it is known that

(i) $\lim_{x \rightarrow \pm\infty} f(x) = +\infty$

(ii) $\lim_{x \rightarrow -1} f(x) = 6$

(iii) $\lim_{x \rightarrow -1} f(x) = \text{a non-zero finite number}$

Let L and M denote the number of points of discontinuity and the number of points of non-differentiability respectively of any function in R. (R is the set of real numbers)

मान लीजिए $f(x) = \frac{(x^2 - 1)(x^2 - 4)}{g(x)}$, $x \in R - \{a, b\}$, जहाँ $g(x)$ कोटि ≤ 4 का एक बहुपद है। $f(x)$ $x = a$ तथा $x =$

b पर अपनेय असततता रखता है। दिया गया है

(i) $\lim_{x \rightarrow \pm\infty} f(x) = +\infty$

(ii) $\lim_{x \rightarrow -1} f(x) = 6$

(iii) $\lim_{x \rightarrow -1} f(x) = \text{एक अशून्य परिमित संख्या}$

माना L तथा M किसी फलन R में क्रमशः असततता बिन्दुओं तथा अन-अवकलनीयता बिन्दुओं की संख्या को दर्शाता है। (R वास्तविक संख्याओं का समुच्चय है)

D05/3/139. Number of points where $|f(x)|$ is non-differentiable in its domain is :

बिन्दुओं की संख्या जहाँ $|f(x)|$ अपने प्रांत में अन-अवकलनीय है :

(A) 2 (B) 3 (C) 4 (D) 5

Ans. B

Sol.

D05/3/140. Number of points where $f(|x|)$ is non-differentiable in its domain is :

बिन्दुओं की संख्या जहाँ $f(|x|)$ अपने प्रांत में अन-अवकलनीय है :

(A) 2 (B) 3 (C) 4 (D) 5

Ans. C

Sol.

D05/3/141. Number of points where $f(-|x|)$ is non-differentiable in its domain is :

बिन्दुओं की संख्या जहाँ $f(-|x|)$ अपने प्रांत में अन-अवकलनीय है :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol. Given $\lim_{x \rightarrow \pm\infty} f(x) = \infty$

$\therefore$ degree of $g(x)$ must be less than 4.

Also $\lim_{x \rightarrow -1} f(x) = 6$

$$\therefore \lim_{x \rightarrow -1} \frac{(x^2 - 1)(x^2 - 4)}{g(x)} = 6$$

$\therefore g(x)$ must have $(x + 1)$ as a factor.

Also $\lim_{x \rightarrow -2} f(x) \neq 0$ but finite

$\Rightarrow g(x)$ must have $(x + 2)$ another factor.

$$\Rightarrow g(x) = k(x + 1)(x + 2)$$

$$\Rightarrow \lim_{x \rightarrow -1} \frac{(x^2 - 1)(x^2 - 4)}{k(x + 1)(x + 2)} = 6$$

$$\Rightarrow \lim_{x \rightarrow -1} \frac{(x - 1)(x - 2)}{k} = 6$$

Thus $f(x) = \frac{(x^2 - 1)(x^2 - 4)}{g(x)}$

$$= \frac{(x^2 - 1)(x^2 - 4)}{(x + 1)(x + 2)}, x \neq -1, -2$$

$$= (x - 1)(x - 2), x \neq -1, -2$$

$|f(x)| = |(x - 1)(x - 2)|$ is non-differentiable at $x = 1$ and $x = 2$

$f(|x|) = (|x| - 1)(|x| - 2)$ is non-differentiable at $|x| = 1, 2$ and $x = 0$

Or $x = \pm 1, \pm 2, 0$

$f(-|x|) = (-|x| - 1)(-|x| - 2) = (|x| + 1)(|x| + 2)$ is non-differentiable at $x = 0$ only.

Paragraph D05/3/142 to D05/3/145

Let

$$f(x) = \begin{cases} \frac{1}{e^{x-1} - 2}, & x \neq 1 \\ \frac{1}{e^{x-1} + 2}, & x = 1 \end{cases} \quad \text{and} \quad g(x) = \begin{cases} a_n + \sin \pi x, & x \in [2n, 2n + 1] \\ b_n + \cos \pi x, & x \in (2n - 1, 2n) \end{cases}; n \in \mathbb{Z}.$$

Now, based on above information, answer the following questions

माना

$$f(x) = \begin{cases} \frac{1}{e^{x-1} - 2}, & x \neq 1 \\ 1, & x = 1 \end{cases} \quad \text{तथा } g(x) = \begin{cases} a_n + \sin \pi x, & x \in [2n, 2n+1] \\ b_n + \cos \pi x, & x \in (2n-1, 2n) \end{cases}; n \in \mathbb{Z}.$$

अब, दी गई जानकारी के आधार पर नीचे दिए गए प्रश्नों के उत्तर दीजिए

D05/3/142. Which of the following options is/are correct ?

निम्न में से कौनसा/से विकल्प सही होगा/होंगे ?

(A) $\lim_{x \rightarrow 1^+} f(x) = 1$ (B) $\lim_{x \rightarrow 1^-} f(x) = -1$ (C) $\lim_{x \rightarrow 1} f(x) = 1$ (D) $\lim_{x \rightarrow 1} f(x) = -1$

Ans. AB

Sol.

D05/3/143. If g is continuous for all $x \in \mathbb{R}$, then :

यदि $g \forall x \in \mathbb{R}$ में सतत् है, तब :

(A) $a_{n-1} = b_{n-1}$ (B) $a_n - b_n = 1$ (C) $a_n - b_{n+1} = 1$ (D) $a_{n-1} - b_n = -1$

Ans. BD

Sol.

D05/3/144. The value of $b + p + |q|$ is equal to :

$b + p + |q|$ का मान बराबर होगा :

(A) (B) (C) (D)

Ans. 1.33

Sol.

D05/3/145. The value of $\lambda - 2g'(0)$ is equal to :

$\lambda - 2g'(0)$ का मान बराबर होगा :

(A) (B) (C) (D)

Ans. 2

Sol.

Paragraph for Question D02/3/146 to D02/3/148

The function $f(x) = mx$ satisfies $f(x + y) = f(x) + f(y)$ and $f(x) = a^x$ satisfies $f(x + y) = f(x)f(y)$.

From the given functional relations, we can determine several things about the functions. At times the function can be determined uniquely from the functional equation.

D02/3/146. If $f(x + y) = f(x) + f(y)$ for all x, y then $f(4)$ is equal to :

- (A) $f(1)$ (B) $4f(1)$ (C) $2f(1)$ (D) 0

Ans. B

Sol.

D02/3/147. If $f(x + y) = f(x) + f(y)$ for all x, y and $f(1) = 1$ then $f(-9/8)$ is equal to :

- (A) $9/8$ (B) $8/9$ (C) $-9/8$ (D) 1

Ans. C

Sol.

D02/3/148. If $f(x) + f(y) = f\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$ then :

- (A) $f(4x^2 + 3x) + 3f(x) = 0$ (B) $f(3x - 4x^3) + 3f(x) = 0$
(C) $f(4x^3 + 3x) - 3f(x) = 0$ (D) $f(4x^3 - 3x) + 3f(x) = 0$

Ans. D

Sol. Putting $x = 1 = y$ in $f(x + y) = f(x) + f(y)$, we obtain $f(2) = 2f(1)$. Assuming that $f(k) = kf(1)$. $f(k + 1) = f(k) + f(1) = kf(1) + f(1) = (k + 1)f(1)$. Thus by induction. $f(n) = nf(1)$ for all $n \in \mathbb{N}$. If $n = -m$, $m \in \mathbb{N}$ then $-f(m) = (-m)f(1)$ but $f(n + (-n)) = f(n) + f(-n)$ and $f(0) = 0$ so $f(-n) = -f(n)$. Thus $f(n) = nf(1)$ for all $n \in \mathbb{I}$. If

$x = p/q$, $p, q \in \mathbb{I}$, $q > 0$ then $f(p) = f(qx)$ but $f(p) = p f(1)$ and $f(qx) = \underbrace{f(x + \dots + x)}_{q \text{ times}} = qf(x)$ so

$f(x) = \frac{p}{q} f(1)$. Thus $f(4) = 4f(1)$ and $f(-9/8) = -9/8 f(1)$.

For Q. 3 putting, $x = y = 0$, we obtain $f(0) = 0$. Putting $y = x$, we get $2f(x) = f\left(2x\sqrt{1-x^2}\right)$. Also replacing

y by $2x\sqrt{1-x^2}$, we get $f(x) + f\left(2x\sqrt{1-x^2}\right) = f(3x - 4x^3) = -f(4x^3 - 3x)$ (putting $y = -x$; we can see that f is an odd function)

$$\Rightarrow 3f(x) = -f(4x^3 - 3x) \Rightarrow f(4x^3 - 3x) + 3f(x) = 0$$

Paragraph for Question D02/3/149 to D02/3/151

$f(x)$ and $g(x)$ are two functions defined $\forall x \in \mathbb{R}$. Point A ($a, f(a)$) is chosen on the graph of $y = f(x)$. A line parallel to x -axis is drawn from point A meeting the bisector of first and third quadrant in Q. Another line parallel to y -axis, from Q, meets the graph of $y = g(x)$ at B. Now a line is drawn parallel to x -axis from point B and yet another line is drawn parallel to y -axis from point A. These two lines meet at a point P.

D02/3/149. If $a = 2$, $f(x) = 2x + 1$, $g(x) = 3x - 1$, coordinate of point B is :

- (A) (14,5) (B) (5,11) (C) (11,5) (D) (5,14)

Ans. D

Sol.

D02/3/150. Locus of point P is :

- (A) $y = f(g(x))$ (B) $y = x + g(x)$ (C) $y = g(f(x))$ (D) $y = x + f(x)$

Ans. C

Sol.

D02/3/151. The curve determined by locus of $y = \frac{f(x)}{g(x)}$ is ($f(x)$ & $g(x)$ are same as in Q.013) :

- (A) Rectangular hyperbola (B) Parabola
(C) Pair of straight line (D) Circle

Ans. A

Sol.

Paragraph for question (D03/3/152–D03/3/153)

Consider $f(x) = \cos(\tan^{-1}x + \cot^{-1}x - x)$, $g(x) = \cos(\sin^{-1}x + \cos^{-1}x - x)$,

$h(x) = \cos(\sec^{-1}x + \operatorname{cosec}^{-1}x - x)$

D03/3/152. Number of periodic function(s) among $f(x)$, $g(x)$ & $h(x)$ is (are) equal to :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol.

D03/3/153. Range of the function $f(x) + g(x) + h(x)$:

- (A) contains one element (B) contains 2 elements
(C) Sum of all the elements of range is $6 \sin 1$ (D) Sum of all the elements of range is 0

Ans. BD

Sol.

Paragraph for question (D03/3/154–D03/3/155)

Let sets A and B are defined as

$$A = \left\{ k \in \mathbb{R} \mid \text{Function } f : \mathbb{R} \rightarrow \left(0, \frac{\pi}{3} \right]; f(x) = \cos^{-1} \frac{x^2 - k}{x^2 + 1} \text{ is Surjective} \right\} \text{ and}$$

$$B = \left\{ x \in \mathbb{Z} \mid x = \csc^{-1} \left(a + \frac{1}{a} \right) + \sec^{-1} \left(b + \frac{1}{b} \right); a, b \in \mathbb{R}, ab < 0 \right\}.$$

Now, based on above information, answer the following questions

D03/3/154. The number of elements in set A is equal to :

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. A

Sol.

D03/3/155. The number of elements in set B is equal to :

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol.

Paragraph for Question Nos. 156 to 158

Consider three function $f(x) = x^2 - 4x$; $g(x) = x + 5$, $h(x) = x - 12$.

On the basis of above information, answer the following questions :

फलनों $f(x) = x^2 - 4x$; $g(x) = x + 5$, $h(x) = x - 12$ पर विचार करें।

दी गई जानकारी के आधार पर निम्नलिखित प्रश्नों के उत्तर दीजिए :

D04/3/156. Number of integral values, which are not in the range of $\frac{g(f(x))}{h(f(x))}$ is :

- (A) one (B) two (C) three (D) four

पूर्णांक मानों की संख्या जो $\frac{g(f(x))}{h(f(x))}$ के परिसर में नहीं हैं :

- (A) एक (B) दो (C) तीन (D) चार

Ans. B

Sol.
$$\frac{g(f(x))}{h(f(x))} = y = \frac{x^2 - 4x + 5}{x^2 - 4x - 12}$$

$$\Rightarrow (y - 1)x^2 + (4 - 4y)x - (12y + 5) = 0$$

$$D \geq 0$$

$$\Rightarrow (4 - 4y)^2 + 4(y - 1)(12y + 5) \geq 0$$

$$\Rightarrow 16y^2 - 15y - 1 \geq 0$$

$$(y - 1)(16y + 1) \geq 0$$



But at $y = 1$

$$1 = \frac{x^2 - 4x + 5}{x^2 - 4x - 12}$$

$$\Rightarrow -12 = 5 \text{ Not possible}$$

$$\therefore \text{Range of } y : \left(-\infty, -\frac{1}{16}\right] \cup (1, \infty)$$

$$\therefore \text{Integral values : } \{0, 1\}$$

D04/3/157. If the largest interval of x for which $g(f(x))$ is invertible is taken as $[a, \infty)$ and least value of $h(f(x))$ is 'b' then

$$\text{the value of } \sin^{-1}\left(\frac{8}{b}\right) + \cos^{-1}\left(-\frac{a^3}{b}\right) + \tan^{-1}\left(-\frac{32}{ab}\right) \text{ is :}$$

यदि x का सबसे बड़ा अंतराल जिसके लिए $g(f(x))$ व्युत्क्रमणीय है, को $[a, \infty)$ के रूप में लिखा जाता है तथा $h(f(x))$ का न्यूनतम

$$\text{मान 'b' है तब } \sin^{-1}\left(\frac{8}{b}\right) + \cos^{-1}\left(-\frac{a^3}{b}\right) + \tan^{-1}\left(-\frac{32}{ab}\right) \text{ का मान होगा :}$$

$$(A) \frac{2\pi}{3}$$

$$(B) \frac{3\pi}{3}$$

$$(C) \frac{5\pi}{12}$$

$$(D) \pi$$

Ans. C

Sol. $g(f(x)) = x^2 - 4x + 5$

$$\text{Largest interval for invertible : } \left[-\frac{B}{2A}, \infty\right)$$

$$\therefore a = \frac{-(-4)}{2.1} = 2$$

$$\text{Least value of } h(f(x)) = x^2 - 4x - 12$$

$$b = -\frac{(16 + 48)}{4} = -16$$

$$\begin{aligned}
& \therefore \sin^{-1}\left(\frac{8}{-16}\right) + \cos^{-1}\left(\frac{-8}{-16}\right) + \tan^{-1}\left(\frac{-32}{2 \times (-16)}\right) \\
& = \sin^{-1}\left(-\frac{1}{2}\right) + \cos^{-1}\left(\frac{1}{2}\right) + \tan^{-1}(1) \\
& = -\frac{\pi}{6} + \frac{\pi}{3} + \frac{\pi}{4} = \frac{\pi}{4} + \frac{\pi}{6} = \frac{5\pi}{12}
\end{aligned}$$

D04/3/158. Value of $\lim_{x \rightarrow 7} \lim_{n \rightarrow \infty} \frac{[g(h(x))]^{2n} + 1}{h(f(x))}$ is equal to :

- (A) $\frac{4}{9}$ (B) $\frac{2}{9}$ (C) $\frac{1}{9}$ (D) Limit does not exist

$\lim_{x \rightarrow 7} \lim_{n \rightarrow \infty} \frac{[g(h(x))]^{2n} + 1}{h(f(x))}$ का मान बराबर है :

- (A) $\frac{4}{9}$ (B) $\frac{2}{9}$ (C) $\frac{1}{9}$ (D) विद्यमान नहीं है

Ans. C

Sol. $\lim_{x \rightarrow 7} \lim_{n \rightarrow \infty} \frac{(x-7)^{2n} + 1}{x^2 - 4x - 12}$

$$\begin{aligned}
\lim_{x \rightarrow 7} \lim_{n \rightarrow \infty} \frac{((x-7)^2)^n + 1}{x^2 - 4x - 12} &= \lim_{x \rightarrow 7} \frac{1}{(x-6)(x+2)} \\
&= \frac{1}{1 \cdot 9} = \frac{1}{9}
\end{aligned}$$

Paragraph for Question Nos. 159 to 160

$$\text{Let } f(x) = \lim_{n \rightarrow \infty} \frac{x^2 + 2(x+1)^{2n}}{(x+1)^{2n+1} + x^2 + 1}, n \in \mathbb{N} \text{ and } g(x) = \tan \left(\frac{1}{2} \sin^{-1} \left\{ \frac{2f(x)}{1+f^2(x)} \right\} \right)$$

माना $f(x) = \lim_{n \rightarrow \infty} \frac{x^2 + 2(x+1)^{2n}}{(x+1)^{2n+1} + x^2 + 1}$, $n \in \mathbb{N}$ तथा $g(x) = \tan \left(\frac{1}{2} \sin^{-1} \left\{ \frac{2f(x)}{1+f^2(x)} \right\} \right)$

D04/3/159. If $x \in (-2, 0)$ then range of $g(x)$ is :

यदि $x \in (-2, 0)$ है, तो $g(x)$ का परिसर होगा :

(A) $\left(0, \frac{4}{5}\right)$ (B) $\left(0, \frac{40}{9}\right)$ (C) $\left(0, \frac{5}{4}\right)$ (D) $\left(0, \frac{25}{16}\right)$

Ans. A

Sol. If $x \in (-2, 0)$ then $x+1 \in (-1, 1) \Rightarrow (x+1)^2 \in [0, 1)$

$$\therefore (x+1)^{2n} \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\therefore f(x) = \lim_{n \rightarrow \infty} \frac{x^2 + 2(x+1)^{2n}}{(x+1)^{2n+1} + x^2 + 1} = \frac{x^2}{x^2 + 1}$$

$$\Rightarrow f(x) = 1 - \frac{1}{x^2 + 1}$$

$$f(x) \in \left(0, \frac{4}{5}\right) \subset \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \text{ as } x \in (-2, 0)$$

$$f(x) = \begin{cases} \frac{x^2}{x^2 + 1}; & x \in (-2, 0) \\ \frac{2}{x+1}; & (-\infty, -2) \cup (0, \infty); f(x) = \tan \theta \therefore \theta = \tan^{-1}(f(x)) \\ 1; & x = 0 \\ 3; & x = -2 \end{cases}$$

$$\Rightarrow g(x) = \tan \left(\frac{1}{2} \sin^{-1} \left(\frac{2f(x)}{1+f^2(x)} \right) \right)$$

$$\Rightarrow g(x) = \tan \left(\frac{1}{2} 2 \tan^{-1} f(x) \right) \text{ as } f(x) \in \left(0, \frac{4}{5}\right) = f(x)$$

$$\text{So, range of } g(x) \text{ is } \left(0, \frac{4}{5}\right)$$

D04/3/160. $\lim_{x \rightarrow -3} \frac{\sin(x+3) \cdot g(x)}{x^2 + 4x + 3}$ is equal to :

- (A) 0 (B) -2 (C) $\frac{1}{2}$ (D) non-existent

$\lim_{x \rightarrow -3} \frac{\sin(x+3) \cdot g(x)}{x^2 + 4x + 3}$ का मान होगा :

- (A) 0 (B) -2 (C) $\frac{1}{2}$ (D) अस्तित्व में नहीं

Ans. C

Sol. If $x < -2$ then $x + 1 < -1$ and $(x + 1)^{2n} \rightarrow \infty$ as $n \rightarrow \infty$

$$\therefore f(x) = \lim_{n \rightarrow \infty} \frac{x^2 + 2(x+1)^{2n}}{(x+1)^{2n+1} + x^2 + 1}$$

$$= \lim_{n \rightarrow \infty} \frac{\frac{x^2}{(x+1)^{2n}} + 2}{x+1 + \frac{x^2 + 1}{(x+1)^{2n}}} = \frac{2}{x+1}$$

$f(x) \in (-2, 0)$ as $x \in (-\infty, -2)$

If $x \in (-\infty, -3]$, then $f(x) \in [-1, 0)$

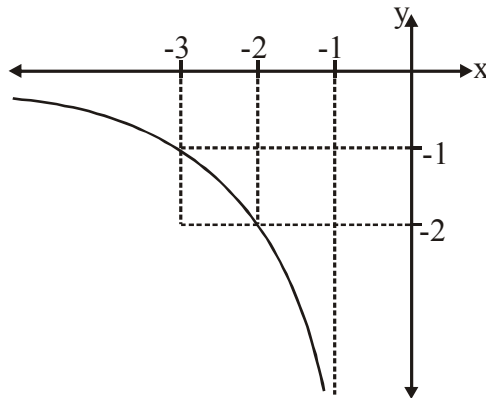
$$\text{So, } g(x) = \tan\left(\frac{1}{2} \cdot 2 \tan^{-1} f(x)\right) = f(x) = \frac{2}{x+1}$$

If $x \in (-3, -2)$ then $f(x) \in (-2, -1)$

$$\therefore g(x) = \tan\left(\frac{1}{2}(-\pi - 2 \tan^{-1} f(x))\right)$$

$$= -\tan\left(\frac{\pi}{2} + \tan^{-1} f(x)\right) = \cot(\tan^{-1} f(x))$$

$$= \cot\left(-\pi + \cot^{-1} \frac{1}{f(x)}\right) = \frac{1}{f(x)} = \frac{x+1}{2}$$



Now, $\lim_{x \rightarrow -3} \frac{\sin(x+3)g(x)}{x^2 + 4x + 3}$

$$\Rightarrow \lim_{x \rightarrow -3^-} \frac{\sin(x+3)}{(x+3)(x+1)} \cdot \frac{2}{(x+1)} = \frac{2}{4} = \frac{1}{2}$$

$$\text{and } \lim_{x \rightarrow -3^+} \frac{\sin(x+3)}{(x+3)(x+1)} \cdot \frac{x+1}{2} = \frac{1}{2}$$

Paragraph for Question Nos. 161 to 163

$$A_i = \frac{x - a_i}{|x - a_i|}, i = 1, 2, \dots, n, \text{ and } a_1 < a_2 < a_3 < \dots < a_n.$$

$$A_i = \frac{x - a_i}{|x - a_i|}, i = 1, 2, \dots, n, \text{ तथा } a_1 < a_2 < a_3 < \dots < a_n.$$

D04/3/0161. If $1 \leq m \leq n$, $m \in \mathbb{N}$, then the value of $L = \lim_{x \rightarrow a_m^-} (A_1 A_2 \dots A_n)$ is :

- (A) always 1 (B) always -1 (C) $(-1)^{n-m+1}$ (D) $(-1)^{n-m}$

यदि $1 \leq m \leq n$, $m \in \mathbb{N}$, तो $L = \lim_{x \rightarrow a_m^-} (A_1 A_2 \dots A_n)$ का मान होगा :

- (A) हमेशा 1 (B) हमेशा -1 (C) $(-1)^{n-m+1}$ (D) $(-1)^{n-m}$

Ans. C

Sol.

D04/3/162. If $1 \leq m \leq n$, $m \in \mathbb{N}$, then the value of $L = \lim_{x \rightarrow a_m^+} (A_1 A_2 \dots A_n)$ is :

- (A) always 1 (B) always -1 (C) $(-1)^{m+1}$ (D) $(-1)^{n-m}$

यदि $1 \leq m \leq n$, $m \in \mathbb{N}$, तो $L = \lim_{x \rightarrow a_m^+} (A_1 A_2 \dots A_n)$ का मान है :

- (A) हमेशा 1 (B) हमेशा -1 (C) $(-1)^{m+1}$ (D) $(-1)^{n-m}$

Ans. D

Sol.

D04/3/163. If $1 \leq m \leq n$, $m \in \mathbb{N}$, then $L = \lim_{x \rightarrow a_m^+} (A_1 A_2 \dots A_n)$

(A) is always equal to -1

(B) is always equal to $+1$

(C) does not exist

(D) None of these

यदि $1 \leq m \leq n$, $m \in \mathbb{N}$, तब $L = \lim_{x \rightarrow a_m^+} (A_1 A_2 \dots A_n)$

(A) हमेशा -1 के बराबर

(B) हमेशा $+1$ के बराबर

(C) विद्यमान नहीं है

(D) इनमें से कोई नहीं

Ans.

C

Sol.

Q.78 - Q.80

We have $A_i = \frac{x - a_i}{|x - a_i|}$, $i = 1, 2, \dots, n$,

And $a_1 < a_2 < \dots < a_{n-1} < a_n$.

Let x be in the left neighbourhood of a_m .

Then $x - a_i < 0$ for $i = m, m+1, \dots, n$

And $x - a_i > 0$ for $i = 1, 2, \dots, m-1$

And $A_i = \frac{x - a_i}{-(x - a_i)} = -1$ for $i = m, m+1, \dots, n$

$A_i = \frac{x - a_i}{x - a_i} = 1$ for $i = 1, 2, \dots, m-1$

Similarly, if x is in the right neighbourhood of a_m , then $x - a_i < 0$ for $i = m+1, \dots, n$, and $x - a_i > 0$ for $i = 1, 2, \dots, m$.

Therefore,

$A_i = \frac{x - a_i}{-(x - a_i)} = -1$ for $i = m+1, \dots, n$

$\lim_{x \rightarrow a_m^-} (A_1 A_2 \dots A_n) = (-1)^{n-m+1}$

And $\lim_{x \rightarrow a_m^+} (A_1 A_2 \dots A_n) = (-1)^{n-m}$

Hence, $\lim_{x \rightarrow a_m} (A_1 A_2 \dots A_n)$ does not exist.

Paragraph for Question D09/3/164 to D09/3/165

Consider a cone of radius R , height H and slant height L . This conical vessel is completely filled with water. A sphere is immersed in this conical vessel. Volume of water displaced by sphere from the cone is equal to volume of the sphere immersed in the cone. The sphere is immersed till it touches the cone.

माना कि त्रिज्या R , ऊँचाई H और तिर्यक ऊँचाई L का एक शंकु है। यह शंकु नुमा पात्र पूरी तरह है पानी से भरा हुआ है। एक गोला इस शंकु पात्र में डाला जाता है। गोले के द्वारा शंकु से हटाये गये पानी का आयतन, शंकु में डाले गये गोले के आयतन के

बराबर है। गोला इस तरह से डुबोया गया है कि यह शंकु को स्पर्श करता है।

D09/3/164. If sphere is immersed upto depth h , what volume of water it has displaced, given that, it's radius is r :

यदि गोले को गहराई h , तक डुबोया जाता है तब हटाये गये पानी का आयतन क्या होगा ? जबकि इसकी त्रिज्या r है :

- (A) $\frac{\pi h^2}{3}(3r-h)$ (B) $\frac{\pi h^2}{3}(2r-h)$ (C) $\frac{\pi h^2}{3}(3r-2h)$ (D) $\frac{\pi r^2}{3}(3r-h)$

Ans. A

Sol.

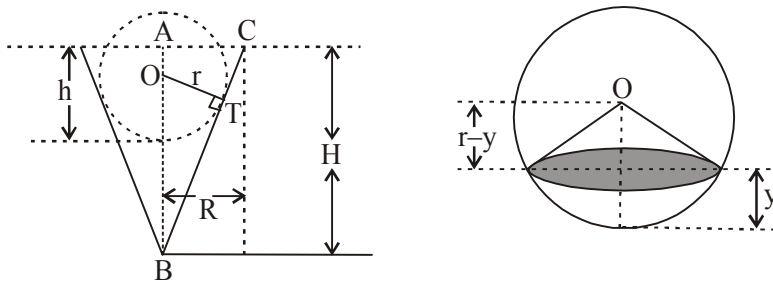
D09/3/165. The value of radius of sphere for which volume displaced by it is maximum :

गोले की त्रिज्या का मान होगा जबकि इसके द्वारा हटाये गये पानी का आयतन अधिकतम है :

- (A) $\frac{2RH}{L+R}$ (B) $\frac{RHL}{(L-R)(L+R)}$ (C) $\frac{RHL}{(L-R)(L+2R)}$ (D) $\frac{2RH}{L+2R}$

Ans. C

Sol.



O is centre of sphere

$$\text{radius of shded disc} = r' = \sqrt{r^2 - (r-y)^2} = \sqrt{2ry - y^2}$$

$$\text{Immersed volume as function of "h" (immersed height)} = \int_0^h \pi(2ry - y^2) dy = \pi r^2 h - \frac{\pi h^3}{3} = \frac{\pi h^2}{3}(3r-h)$$

Note that ABC and OBT are similar triangles

$$\frac{R}{r} = \frac{L}{OB} = \frac{H}{BT}, \text{ now } OB = H - h + r$$

$$\therefore \frac{R}{r} = \frac{L}{H-h+r} \Rightarrow RH - Rh + Rr = rL \Rightarrow r = \frac{R(H-h)}{L-R}$$

$$\text{Hence } V(h) = \pi \left\{ \frac{R(H-h)}{L-R} h^2 - \frac{h^3}{3} \right\}$$

$$\frac{dV}{dh} = \pi \left\{ \frac{2RHh - 3Rh^2}{L - R} - h^2 \right\} = 0 \quad \Rightarrow h = \frac{2RH}{L + 2R}$$

$$\therefore r = \frac{R}{L - R} \left(H - \frac{2RH}{L + 2R} \right) = \frac{RHL}{(L - R)(L + 2R)}$$

Paragraph for Question D09/3/166 to D09/3/168

Consider a function whose derivatives upto n order exists $\forall n \in \mathbb{N}$. If $f(\alpha) = 0$ and $f'(\alpha) \neq 0$, then $x = \alpha$ is called a simple root of $f(x) = 0$ or a root of multiplicity 1. In general if $f(\alpha) = 0, f_1(\alpha) = f_2(\alpha) = \dots, f_{k-1}(\alpha) = 0$ and $f_k(\alpha) \neq 0$, then $x = \alpha$ is said to be the root of $f(x) = 0$ with multiplicity K , where $f_r(x)$ represents the derivative of order r .

If $f(\alpha) = g(\alpha), f'(\alpha) = g'(\alpha)$ and $f''(\alpha) = g''(\alpha)$ then $f(x)$ and $g(x)$ have the same circle of curvature whose

$$\text{radius, } \rho = \left| \frac{\left\{ 1 + \left\{ f'(\alpha)^2 \right\}^{\frac{3}{2}} \right\}}{f''(\alpha)} \right|. \text{ Centre of the circle of curvature lies on the normal at the point at distance } \rho$$

from point of tangency P . It lies below P if curve is concave downwards and above P if curve is concave upwards.

एक फलन $f(x)$ जिसके $n \in \mathbb{N}$ क्रम तक के अवकलज के अस्तित्व हैं। यदि $f(\alpha) = 0$ तथा $f'(\alpha) \neq 0$, तो $x = \alpha$ समीकरण $f(x) = 0$ का एक साधारण मूल है या इकाई पुरावृत्तिता (multiplicity) वाला मूल है। व्यापक रूप से यदि $f(\alpha) = 0, f_1(\alpha) = f_2(\alpha) = \dots, f_{k-1}(\alpha) = 0$ तथा $f_k(\alpha) \neq 0$, तो $x = \alpha$ समीकरण $f(x) = 0$ का K पुनरावृत्तिता (multiplicity) वाला मूल होगा जहाँ $f_r(x)$, r क्रम के अवकलज को बताता है।

यदि $f(\alpha) = g(\alpha), f'(\alpha) = g'(\alpha)$ तथा $f''(\alpha) = g''(\alpha)$ तो $f(x)$ तथा $g(x)$ समान वक्रता वृत्त रखते हैं जिसकी त्रिज्या

$$\rho = \left| \frac{\left\{ 1 + \left\{ f'(\alpha)^2 \right\}^{\frac{3}{2}} \right\}}{f''(\alpha)} \right| \text{ है। वक्रता वृत्त का केन्द्र, स्पर्श बिन्दु } P \text{ पर खींचे गए अभिलम्ब पर } \rho \text{ दूरी पर स्थित बिन्दु होता है। यह}$$

बिन्दु P से नीचे की ओर होता है। यदि वक्र नीचे की तरफ अवतलीय (concave downwards) तथा ऊपर होगा यदि ऊपर की ओर अवतलीय हो।

D09/3/166. $f(x) = \int_0^x (e^t - 1)t^2(t-2)^2(t-3)dt - \frac{x^4}{4}$, $x=0$ is a root of equation $f'(x)$ with multiplicity :

$$f(x) = \int_0^x (e^t - 1)t^2(t-2)^2(t-3)dt - \frac{x^4}{4}, \text{ समीकरण } f'(x) = 0 \text{ के लिए } x = 0 \text{ मूल की पुनरावृत्ति कितनी बार होगी}$$

:

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. C

Sol. $f'(x) = x^2(e^x - 1)(x-2)^2(x-3) - x^3$

$$= \frac{x^2}{2} \left\{ \underbrace{(e^x - 1)(x-2)^2(x-3)}_1 \right\} - x^3$$

Hence $x = 0$ is a root of multiplicity 3

Hence (C)

D09/3/167. If $f(x)$ and $g(x)$ have same circle of curvature at $x = \alpha$ and $\lim_{x \rightarrow \alpha^+} \frac{f(x) - g(x)}{(x - \alpha)^3} = 0$, then $x = \alpha$ is a root of $f(x) - g(x)$

$-g(x) = 0$ whose multiplicity is atleast :

यदि $x = \alpha$ पर $f(x)$ तथा $g(x)$ समान वक्रता व त रखते हो तथा $\lim_{x \rightarrow \alpha^+} \frac{f(x) - g(x)}{(x - \alpha)^3} = 0$, तो $x = \alpha$ समीकरण $f(x) - g(x)$

$= 0$ का मूल होगा जिसकी पुनरावृत्ति कम से कम निम्न होगी :

- (A) 2 (B) 3 (C) 4 (D) 5

Ans. C

Sol. $\lim_{x \rightarrow \alpha^+} \frac{f(x) - g(x)}{(x - \alpha)^3} = \lim_{x \rightarrow \alpha^+} \frac{f'(x) - g'(x)}{3(x - \alpha)^2} = \lim_{x \rightarrow \alpha^+} \frac{f''(x) - g''(x)}{6(x - \alpha)}$

$$= \lim_{x \rightarrow \alpha^+} \frac{f'''(x) - g'''(x)}{6} = 0 \Rightarrow f'''(\alpha) = g'''(\alpha)$$

Hence $x = \alpha$ is a root of multiplicity atleast 4.

Hence (C)

D09/3/168. $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3 - cx$ ($c > 0$). The circle of curvature of $f(x)$ at the point of extremum has its centre on x -axis, then c is :

यदि $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3 - cx$ ($c > 0$) के चरम बिन्दु पर वक्रता वृत्त का केन्द्र x -अक्ष पर हो, तो c का मान होगा :

- (A) $\sqrt{\frac{3}{2}}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $2\sqrt{3}$

Ans. C

Sol.

$$f'(x) = 3x^2 - C = 0 \begin{cases} x = \sqrt{\frac{c}{3}} = \lambda \\ x = -\sqrt{\frac{c}{3}} \end{cases}$$

$$f''(x) = 6x$$

$$f'(\lambda) = 0$$

$$\rho = \left| \frac{1}{f''(\lambda)} \right| = |f(\lambda)|$$

Hence

$$\frac{1}{6\lambda} = \lambda c - \lambda^3$$

$$\Rightarrow 6c\lambda^2 - 6\lambda^4 = 1$$

$$\Rightarrow 6c \cdot \frac{c}{3} - 6 \cdot \frac{c^2}{9} = 1$$

$$\Rightarrow 2c^2 - \frac{2c^2}{3} = 1$$

$$\Rightarrow c^2 = \frac{3}{4}$$

$$\Rightarrow c = \frac{\sqrt{3}}{2}$$

Hence (C)

Paragraph for Question D09/3/169 to D09/3/170

Let $f(x) = x^3 + ax^2 + bx + c$ be a given cubic polynomial and $f(x) = 0$ be the corresponding cubic equation where $a, b, c \in \mathbb{R}$.

माना $f(x) = x^3 + ax^2 + bx + c$ घनीय बहुपद है तथा $f(x) = 0$ संगत घनीय समीकरण है जहाँ $a, b, c \in \mathbb{R}$ है।

D09/3/169. Which of the following is/are correct ?

(A) If $a^2 - 3b < 0$ and $c > 0$ then $f(x) = 0$ has only one negative real root

(B) If $a^2 - 3b < 0$ and $c < 0$ then $f(x) = 0$ has only one positive real root

(C) If $a^2 - 3b = 0$ then $f(x) = 0$ may have three real and equal roots

(D) If $a^2 - 3b = 0$ then $f(x) = 0$ may have three real roots in which two roots are equal

निम्न में से कौनसा/कौनसे सही है ?

(A) यदि $a^2 - 3b < 0$ और $c > 0$ तब $f(x) = 0$ का केवल एक ऋणात्मक वास्तविक मूल है

(B) यदि $a^2 - 3b < 0$ और $c < 0$ तब $f(x) = 0$ का केवल एक धनात्मक वास्तविक मूल है

(C) यदि $a^2 - 3b = 0$ हो तो $f(x) = 0$ के तीन वास्तविक और बराबर मूल हैं

(D) यदि $a^2 - 3b = 0$ हो तो $f(x) = 0$ के तीन वास्तविक मूल हैं जिसमें से दो मूल बराबर हैं

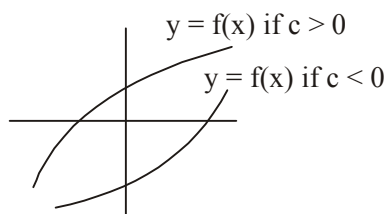
Ans. ABC

Sol. $f'(x) = 3x^2 + 2ax + b$

$$D = 4(a^2 - 3b)$$

If $a^2 - 3b < 0$ then D will be negative and $f'(x) > 0$ for all x . Hence $f(x)$ increasing function for all x .

Now possible graph of $f(x)$ will be



If $a^2 - 3b = 0$ then $D = 0$ hence $f'(x) = 0$ has two equal roots. let x_1

$$f'(x) = 3(x - x_1)^2 \Rightarrow f(x) = (x - x_1)^3 + k$$

$$\text{hence } f(x) = 0 \Rightarrow (x - x_1)^3 = -k$$

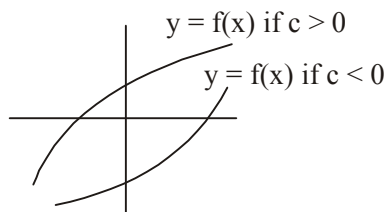
$$\text{when } k = 0 \Rightarrow f(x) = 0 \text{ has all three real and equal roots}$$

$$k \neq 0 \Rightarrow f(x) = 0 \text{ has one real and two imaginary roots}$$

Hindi : $f'(x) = 3x^2 + 2ax + b$

$$D = 4(a^2 - 3b)$$

यदि $a^2 - 3b < 0$ तब D ऋणात्मक होगा तथा $f'(x) > 0$ सभी x के लिए अतः सभी x के लिए, $f(x)$ वर्धमान फलन है
अतः $f(x)$ का संभावित आरेख होगा।



यदि $a^2 - 3b = 0$ हो तब

$$f'(x) = 3(x - x_1)^2 \Rightarrow f(x) = (x - x_1)^3 + k$$

$$\text{अतः } f(x) = 0 \Rightarrow (x - x_1)^3 = -k$$

$$\text{जब } k = 0 \Rightarrow f(x) = 0 \text{ के सभी तीन वास्तविक और बराबर मूल हैं।}$$

$$k \neq 0 \Rightarrow f(x) = 0 \text{ का एक वास्तविक और दो काल्पनिक मूल हैं।}$$

D09/3/170. If $a^2 - 3b > 0$ and x_1 and x_2 are real roots of $f'(x) = 0$ such that $x_2 > x_1$, then which of the following is/are correct ?

- (A) If $f(x_1).f(x_2) > 0$ then $f(x) = 0$ would have just one real root
 (B) If $f(x_1).f(x_2) < 0$ then $f(x) = 0$ would have three real and distinct roots
 (C) If $f(x_1).f(x_2) = 0$ then $f(x) = 0$ would have three real roots but one of which repeated
 (D) $f'(x) < 0 \forall x \in (x_1, x_2)$ and $f'(x) > 0 \forall x \in (-\infty, x_1) \cup (x_2, \infty)$ and $x = x_1$ is point of local minima and $x = x_2$ is point of local max

यदि $a^2 - 3b > 0$ तथा x_1 और x_2 , $f'(x) = 0$ के मूल वास्तविक इस प्रकार हैं कि $x_2 > x_1$, निम्न में से कौनसे सही हैं ?

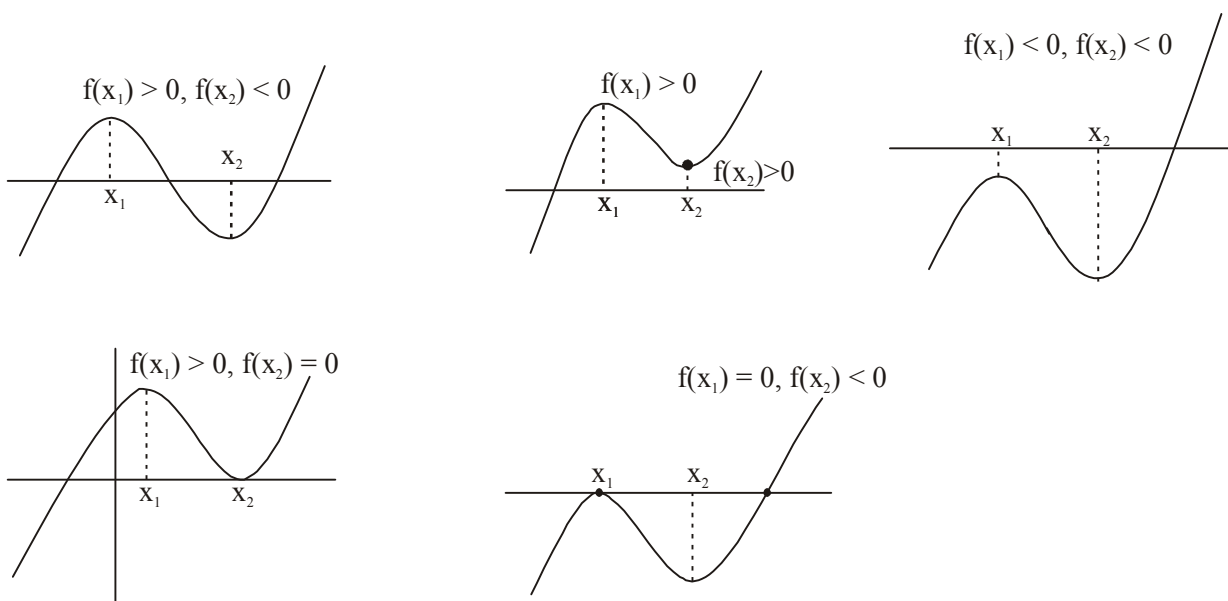
- (A) यदि $f(x_1).f(x_2) > 0$ तब $f(x) = 0$ ठीक एक वास्तविक मूल रखता है
 (B) यदि $f(x_1).f(x_2) < 0$ हो तो $f(x) = 0$ के तीन वास्तविक और असमान मूल हैं
 (C) यदि $f(x_1).f(x_2) = 0$ हो तो $f(x) = 0$ के तीन वास्तविक मूल मूल परंतु उनमें से एक की पुनरावृत्ति है
 (D) $f'(x) < 0 \forall x \in (x_1, x_2)$ और $f'(x) > 0 \forall x \in (-\infty, x_1) \cup (x_2, \infty)$ और $x = x_1$ स्थानीय निम्निष्ठ का बिन्दु $x = x_2$ स्थानीय उच्चिष्ठ का बिन्दु है

Ans. ABC

Sol. $f'(x) = 3x^2 + 2ax + b = 3(x - x_1)(x - x_2) \quad \therefore D = 4(a^2 - 3b) > 0$

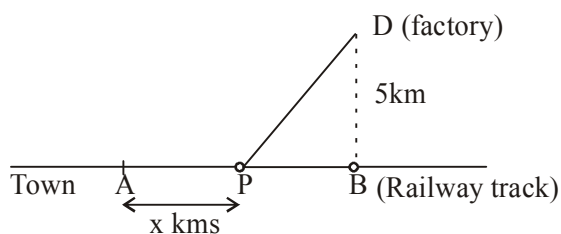
hence अतः $f'(x) > 0 \forall x \in (-\infty, x_1) \cup (x_2, \infty)$ and और $f'(x) < 0 \forall x \in (x_1, x_2)$

now possible graphs are अब संभावित आरेख



Paragraph for Question D09/3/171 to D09/3/172

A factory is connected with a straight railway track by a highway. Town A lies on the railway track. The distance DB from the factory to railway track is 5 km. The segment AB of railway track is equal to ' ℓ ' kms. Freight charges on highway are 2 times higher than on railway. A highway DP is connected to railway track such that AP is x kms (see the figure).



D09/3/171. If ℓ is $\frac{185}{\sqrt{3}}$ kms, then the value of 'x' for which freight charge is minimum :

- (A) $\frac{60}{\sqrt{3}}$ kms (B) $60\sqrt{3}$ kms (C) $120\sqrt{3}$ kms (D) $75\sqrt{3}$ kms

Ans. B

Sol. $f(x) = x + 2\sqrt{25 + (\ell - x)^2}$... (i)

$$0 \leq x \leq \ell$$

$$f'(x) = 1 - \frac{2(\ell - x)}{\sqrt{25 + (\ell - x)^2}} \quad \dots (ii)$$

$$= \frac{\sqrt{25 + (\ell - x)^2} - 2(\ell - x)}{\sqrt{25 + (\ell - x)^2}}$$

$$25 + (\ell - x)^2 = 4(\ell - x)^2$$

$$\sqrt{3}(\ell - x) = \pm 5$$

$$x = \ell \pm \frac{5}{\sqrt{3}}$$

$$x = \ell - \frac{5}{\sqrt{3}}$$

$$\text{If } \ell = \frac{185}{\sqrt{3}} \text{ then } x = \frac{180}{\sqrt{3}} = 60\sqrt{3} \text{ kms}$$

D09/3/172. If $\ell = 10$ kms, then the value of x for which freight charges are maximum :

- (A) 10 kms (B) $10 - \frac{5}{\sqrt{3}}$ kms (C) $\frac{10}{\sqrt{3}}$ kms (D) 0 km

Ans. D

Sol. $f(x) = x + 2\sqrt{25 + (\ell - x)^2}$... (i)

$$0 \leq x \leq \ell$$

$\ell = 10$ kms, then maximum value of $f(x)$ lies at $x = 0$ or $x = 10$

$$f(x) = x + 2\sqrt{25 + (10 - x)^2}$$

$$f(0) = 10\sqrt{5}$$

$$f(10) = 20$$

$$\Rightarrow \text{maximum freight} = 10\sqrt{5} \text{ at } x = 0.$$

Paragraph for Question D09/3/173 to D09/3/175

Consider f, g and h be three real valued differentiable function defined on $\mathbb{R}$.

Let $g(x) = x^3 + g''(1)x^2 + (3g'(1) - g''(1) - 1)x + 3g'(1)$, $f(x) = xg(x) - 12x + 1$ and $f(x) = (h(x))^2$ where $h(0) = 1$.

D09/3/173. The function $y = f(x)$ has :

- (A) Exactly one local minima and no local maxima
- (B) Exactly one local maxima and no local minima
- (C) Exactly one local maxima and two local minima
- (D) Exactly two local maxima and one local minima

Ans. C

Sol. Let $g(x) = x^3 + Ax^2 + Bx + C$
 $\Rightarrow A = g''(1) = 6 + 2A \Rightarrow A = -6$

$$\begin{aligned} B &= 3g'(1) - g''(1) - 1 \\ &= 3g'(1) - g''(1) - 1 \\ &= 3(3 + 2 \times -6 + B) - A - 1 \\ &= 3B - 27 + 5 \end{aligned}$$

$$\Rightarrow B = 11$$

and $C = 3g'(1) = 3(3 + (-12) + 11) = 6$

$$\Rightarrow g(x) = x^3 - 6x^2 + 11x + 6$$

$$\Rightarrow f(x) = x^4 - 6x^3 + 11x^2 - 6x + 1$$

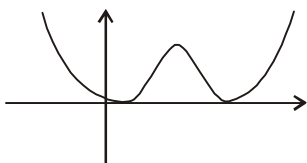
Also $f(x) = (h(x))^2$

$$\Rightarrow h(x) = x^2 - 3x + 1$$

$$f'(x) = 2h(x)h'(x)$$

Graph of $f(x)$ will be of the form

$$\xleftrightarrow[3 \text{ distinct real roots}]{2(x^2 - 3x + 1)(2x - 3)}$$



Graphically or otherwise, $f(x)$ has exactly 1 local maxima and 2 local minima

D09/3/174. Which of the following is(are) true for the function $y = g(x)$?

(A) $g(x)$ monotonically decreases in $\left(-\infty, 2 - \frac{1}{\sqrt{3}}\right) \cup \left(2 + \frac{1}{\sqrt{3}}, \infty\right)$

(B) $g(x)$ monotonically increases in $\left(2 - \frac{1}{\sqrt{3}}, 2 + \frac{1}{\sqrt{3}}\right)$

(C) There exist one tangent to $y = g(x)$ which is parallel to the chord joining the points $(1, g(1))$ and $(3, g(3))$

(D) There exists exactly two distinct Lagrange's mean value in $(0, 4)$ for the function $y = g(x)$

Ans.

D

Sol.

Let $g(x) = x^3 + Ax^2 + Bx + C$

$\Rightarrow A = g''(1) = 6 + 2A \Rightarrow A = -6$

$B = 3g'(1) - g''(1) - 1$

$= 3g'(1) - g''(1) - 1$

$= 3(3 + 2 \times -6 + B) - A - 1$

$= 3B - 27 + 5$

$\Rightarrow B = 11$

and $C = 3g'(1) = 3(3 + (-12) + 11) = 6$

$\Rightarrow g(x) = x^3 - 6x^2 + 11x + 6$

$\Rightarrow f(x) = x^4 - 6x^3 + 11x^2 - 6x + 1$

Also $f(x) = (h(x))^2$

$\Rightarrow h(x) = x^2 - 3x + 1$

$g(x) = (x-1)(x-2)(x-3) + 12$

$g'(x) = 3x^2 - 12x + 11$

Roots are $2 \pm \frac{1}{\sqrt{3}}$

Graphically or otherwise :

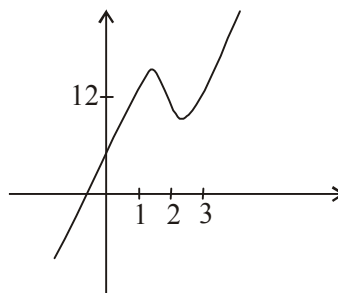
(A) and (B) are clearly incorrect

Also $f(1) = f(3)$

Hence, chord joining $(1, f(1))$ and $(3, f(3))$ is parallel to x-axis and there are 2 points :

$x = 2 \pm \frac{1}{\sqrt{3}}$ where tangents are parallel to x-axis.

$f(0) = 6$



$$f(4) = 18$$

Apply LMVT in $[0, 4]$

$$f'(c) = 3$$

$$\Rightarrow 3c^2 - 12c + 11 = 3$$

$$\Rightarrow 3c^2 - 12c + 8 = 0 \quad \{2 \text{ distinct real roots}\}$$

D09/3/175. Which one of the following does not hold good for $y = h(x)$?

(A) Exactly one critical point

(B) No point of inflection

(C) Exactly one real zero in $(0, 3)$

(D) Exactly one tangent parallel to x-axis

Ans. C

Sol. Let $g(x) = x^3 + Ax^2 + Bx + C$

$$\Rightarrow A = g''(1) = 6 + 2A \Rightarrow A = -6$$

$$B = 3g'(1) - g''(1) - 1$$

$$= 3g'(1) - g''(1) - 1$$

$$= 3(3 + 2 \times -6 + B) - A - 1$$

$$= 3B - 27 + 5$$

$$\Rightarrow B = 11$$

$$\text{and } C = 3g'(1) = 3(3 + (-12) + 11) = 6$$

$$\Rightarrow g(x) = x^3 - 6x^2 + 11x + 6$$

$$\Rightarrow f(x) = x^4 - 6x^3 + 11x^2 - 6x + 1$$

$$\text{Also } f(x) = (h(x))^2$$

$$\Rightarrow h(x) = x^2 - 3x + 1$$

$$h(x) = x^2 - 3x + 1$$

(A), (B) and (D) are true

$$\text{Also } h(0) = h(3) = 1$$

$\Rightarrow$ (C) is incorrect.

Paragraph for Question D09/3/176 to D09/3/178

Let $f(x) = a(x - 2)(x - b)$, where $a, b \in \mathbb{R}$ and $a \neq 0$.

Also $f(f(x)) = a^3 \left(x^2 - (2+b)x + 2b - \frac{2}{a} \right) \left(x^2 - (2+b)x + 2b - \frac{b}{a} \right)$, $a \neq 0$, has exactly one real root as 5.

D09/3/176. Which of the following is correct ?

(A) The minimum value of $f(x)$ is 5 and attained at $x = -2$

(B) The maximum value of $f(x)$ is 2 and attained at $x = 5$

(C) The minimum value of $f(x)$ is 10 and attained at $x = 0$

(D) The minimum value of $f(x)$ is 24 and attained at $x = 6$

Ans. B

Sol. Since $x = 5$ is the only real zero of $f(f(x))$, one of the 2 quadratic factors of $f(f(x))$ will have real and equal roots. {The root being 5} and the other won't have real roots.

Both quadratic factors have $\frac{b}{2} + 1$ as vertex.

$$\Rightarrow \frac{b}{2} + 1 = 5$$

$$\Rightarrow b = 8$$

$$\Rightarrow f(f(x)) = a^3 \left(x^3 - 10x + 16 - \frac{2}{a} \right) \left(x^2 - 10x + 16 - \frac{8}{a} \right)$$

$$\Rightarrow 16 - \frac{2}{a} = 25$$

$$\Downarrow$$

$$16 - \frac{8}{a} = 52$$

or

$$\Rightarrow 16 - \frac{8}{a} = 25$$

$$\Downarrow$$

$$16 - \frac{2}{a} = \frac{73}{4}$$

Other quadratic factor
has no real roots

Other quadratic factor
has no real roots

$$\Rightarrow -\frac{2}{a} = 9$$

$$\Rightarrow a = -\frac{2}{9}$$

$$\Rightarrow f(x) = -\frac{2}{9}(x-2)(x-8)$$

(B) is correct. Rest incorrect.

D09/3/177. The slope of the straight line passing through $O(0,0)$ and tangent to $y = f(x)$, can be :

(A) -4

(B) 2

(C) $\frac{4}{9}$

(D) $\frac{2}{9}$

Ans. C

Sol. Slope of tangent at $(\alpha, f(\alpha)) = \frac{-2}{9}(2\alpha - 10)$

$$\Rightarrow \frac{f(\alpha)}{\alpha} = \frac{-2}{9}(2\alpha - 10) \quad \{\text{tangent is drawn from origin}\}$$

$$\Rightarrow \alpha = \pm 4$$

$$\text{Slope} = \frac{4}{9}, 4$$

D09/3/178. The value of definite integral $\int_2^5 f(x) dx$ is equal to :

- (A) 4 (B) 8 (C) 10 (D) 14

Ans. A

Sol.
$$\int_2^5 f(x) dx = \int_{-3}^0 f(x+5) dx = \int_{-3}^0 \frac{-2}{9}(x^2 - 9) dx = \int_{-3}^0 2 - \frac{2x^2}{9} dx = 4$$

Paragraph for Question D09/3/179 to D09/3/181

Let $f(x)$ be function defined on the set of real numbers to real numbers such it is continuous on $\mathbb{R}$.

D09/3/179. If $f(x)$ is also differentiable on $\mathbb{R}$ such that $f'(x) > f(x) \forall x \in \mathbb{R}$ and $f(x_0) = 0$ then :

- (A) $f(x) < 0 \forall x > x_0$ (B) $f(x) \geq 0 \forall x > x_0$ (C) $f(x) > 0 \forall x > x_0$ (D) $f(x) \leq 0 \forall x > x_0$

Ans. C

Sol. $f'(x) > f(x) \forall x \in \mathbb{R}$

$$\Rightarrow (f'(x) - f(x))e^{-x} > 0 \forall x \in \mathbb{R}$$

$$\Rightarrow \frac{d}{dx}(e^{-x}f(x)) > 0 \forall x \in \mathbb{R}$$

$$\Rightarrow \forall x > x_0 \quad e^{-x_0}f(x_0) < e^{-x}f(x)$$

$$\Rightarrow \forall x > x_0 : f(x)e^{-x} > 0$$

$$\Rightarrow f(x) > 0 \forall x > x_0$$

D09/3/180. The equation $k \cdot e^x = 5 + x + \frac{x^2}{2}$ where k is a positive constant has :

- (A) exactly two roots (B) exactly one root (C) no real root (D) many roots

Ans. B

Sol. Let $f(x) = ke^x - \left(\frac{x^2}{2} + x + 5 \right)$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty$$

$$x \rightarrow \infty, f(x) \rightarrow \infty$$

Exponential function always dominates a polynomial as x increases.

$$\Rightarrow f(x) = 0 \text{ has at least 1 root}$$

Also, $f'(x) = ke^x - (x + 1)$

As $x \rightarrow \pm\infty, f'(x) \rightarrow \infty$

and $f''(x) = ke^x - 1$

$\Rightarrow f''(x)$ has exactly 1 root $-\ln k$

When $x = -\ln k$

$f'(x) = \ln k$

$f'(x)$ has exactly 1 minima at $x = -\ln k$

If $\ln k < 0, \{k \in (0, 1)\}$

$f'(x)$ decreases in $(-\infty, \ln k)$

and increases in $(\ln k, \infty)$

$\Rightarrow f'(x)$ has 2 real roots $\alpha < -\ln k$ and $\beta > -\ln k$

$\Rightarrow f(x)$ increases in $(-\infty, \alpha) \cup (\beta, \infty)$

and decrease ion (α, β)

Also, α, β satisfy : $f'(x) = 0$

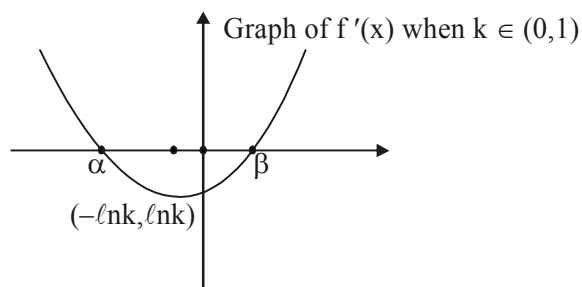
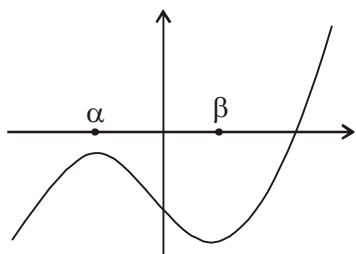
$\Rightarrow ke^\alpha = \alpha + 1$ and $ke^\beta = \beta + 1$

$\Rightarrow f(\alpha) = -\left(\frac{\alpha^2}{2} + 4\right) < 0$ and $f(\beta) = -\left(\frac{\beta^2}{2} + 4\right) < 0$

Hence, graphically or otherwise.

$f(x)$ has exactly 1 root in $\mathbb{R}$.

Graph of $f(x)$



D09/3/181. If $f(x) = x^3 - (9 - a)x^2 + 3(9 - a^2)x + 7$ has points of extrema which are of opposite sign, then the values of parameter a :

(A) $\left(-\infty, -\frac{4}{9}\right) \cup (0, \infty)$

(B) $(-3, -3) \cup (4, \infty)$

(C) $\left(-3, -\frac{4}{9}\right) \cup (0, 3)$

(D) $(-\infty, -3) \cup (3, \infty)$

Ans. D

Sol. $f(x)$ has points of extrema with opposite signs

$\Rightarrow f'(x) = 0$ has roots with opposite signs

$\Rightarrow f'(0) < 0 \Rightarrow 9 - a^2 < 0 \Rightarrow a^2 > 9$

$$\Rightarrow a \in (-\infty, -3) \cup (3, \infty)$$

Paragraph for Question D06/3/182 to D06/3/184

$$\text{Let } f(x) = \begin{cases} \max. \{t^3 - t^2 + t + 1, 0 \leq t \leq x\}, & 0 \leq x \leq 1 \\ \min. \{3 - t, 1 < t \leq x\} & 1 < x \leq 2 \end{cases} \text{ and}$$

$$g(x) = \begin{cases} \max. \left\{ \frac{3}{8}t^4 + \frac{1}{2}t^3 - \frac{3}{2}t^2 + 1, 0 \leq t \leq x \right\}, & 0 \leq x < 1 \\ \min. \left\{ \frac{3}{8}t + \frac{1}{32}\sin^2 \pi t + \frac{5}{8}, 1 \leq t \leq x \right\} & 1 \leq x \leq 2 \end{cases}$$

Define $f(x)$ and $g(x)$ explicitly and then answer the following questions

$$\text{माना } f(x) = \begin{cases} \max. \{t^3 - t^2 + t + 1, 0 \leq t \leq x\}, & 0 \leq x \leq 1 \\ \min. \{3 - t, 1 < t \leq x\} & 1 < x \leq 2 \end{cases} \text{ तथा}$$

$$g(x) = \begin{cases} \max. \left\{ \frac{3}{8}t^4 + \frac{1}{2}t^3 - \frac{3}{2}t^2 + 1, 0 \leq t \leq x \right\}, & 0 \leq x < 1 \\ \min. \left\{ \frac{3}{8}t + \frac{1}{32}\sin^2 \pi t + \frac{5}{8}, 1 \leq t \leq x \right\} & 1 \leq x \leq 2 \end{cases}$$

$f(x)$ तथा $g(x)$ स्पष्ट रूप से परिभाषित हैं तब निम्न प्रश्नों के उत्तर दीजिए :

D06/3/182. The function $f(x)$, $\forall x \in [0, 2]$ is :

- | | |
|--|---------------------------------------|
| (A) continuous and differentiable | (B) continuous but not differentiable |
| (C) discontinuous and not differentiable | (D) none |

फलन $f(x)$, $\forall x \in [0, 2]$ है :

- | | |
|----------------------------|---------------------------|
| (A) सतत् तथा अवकलनीय | (B) सतत् तथा अवकलनीय नहीं |
| (C) असतत् तथा अवकलनीय नहीं | (D) इनमें से कोई नहीं |

Ans. B

Sol. Consider $f(x)$ in $[0, 1]$

$$f'(t) = 3t^2 - 2t + 1 > 0 \forall t \in (0, 1) \Rightarrow f \text{ is } \uparrow \text{ in } (0, 1)$$

maximum occurs at $t = x$.

$$\text{and hence } f(x) \text{ is } \begin{cases} x^3 - x^2 + x + 1 & 0 \leq x \leq 1 \\ 3 - x & 1 < x \leq 2 \end{cases}$$

again consider $g(x)$ in $[0, 1)$

$$g(t) = \frac{3}{8}t^4 + \frac{1}{2}t^3 - \frac{3}{2}t^2 + 1$$

$$g'(t) = \frac{3}{2}t^3 + \frac{3}{2}t^2 - 3t = \frac{3}{2}t(t^2 + t - 2) = \frac{3}{2}t(t-1)(t+2)$$

$g(t)$ decreases in $[0, 1) \Rightarrow$ maximum occurs when $t = 0$ and $g(0) = 1$

again consider $g(x)$ function in $[1, 2]$

$$g'(t) = \frac{3}{8} + \frac{\pi}{32} \sin(2\pi t) > 0 \quad \forall t \in \mathbb{R}$$

$\therefore$ g is an increasing function in $[1, 2] \Rightarrow$ minimum occurs when $t = 0$ and $g(1) = 1$

$$\text{hence } g(x) = \begin{cases} 1 & 0 \leq x < 1 \\ 1 & 1 < x \leq 2 \end{cases} = 1 \quad \forall x \in [0, 2]$$

$f(x)$ is continuous but not differentiable at $x = 1$

D06/3/183. Which of the following is true?

(A) $\lim_{x \rightarrow 1^-} f \circ g(x) > \lim_{x \rightarrow 1^+} g \circ f(x)$

(B) $\lim_{x \rightarrow 1^-} f \circ g(x) < \lim_{x \rightarrow 1^+} g \circ f(x)$

(C) $\lim_{x \rightarrow 1^-} f \circ g(x) = \lim_{x \rightarrow 1^+} g \circ f(x)$

(D) None of these

निम्न में से कौनसा सत्य है?

(A) $\lim_{x \rightarrow 1^-} f \circ g(x) > \lim_{x \rightarrow 1^+} g \circ f(x)$

(B) $\lim_{x \rightarrow 1^-} f \circ g(x) < \lim_{x \rightarrow 1^+} g \circ f(x)$

(C) $\lim_{x \rightarrow 1^-} f \circ g(x) = \lim_{x \rightarrow 1^+} g \circ f(x)$

(D) इनमें से कोई नहीं

Ans. A

Sol. Consider $f(x)$ in $[0, 1]$

$$f'(t) = 3t^2 - 2t + 1 > 0 \quad \forall t \in (0, 1) \Rightarrow f \text{ is } \uparrow \text{ in } (0, 1)$$

maximum occurs at $t = x$.

$$\text{and hence } f(x) \text{ is } \begin{cases} x^3 - x^2 + x + 1 & 0 \leq x \leq 1 \\ 3 - x & 1 < x \leq 2 \end{cases}$$

again consider $g(x)$ in $[0, 1)$

$$g(t) = \frac{3}{8}t^4 + \frac{1}{2}t^3 - \frac{3}{2}t^2 + 1$$

$$g'(t) = \frac{3}{2}t^3 + \frac{3}{2}t^2 - 3t = \frac{3}{2}t(t^2 + t - 2) = \frac{3}{2}t(t-1)(t+2)$$

$g(t)$ = decreases in $[0,1) \Rightarrow$ maximum occurs when $t=0$ and $g(0)=1$

again consider $g(x)$ function in $[1,2]$

$$g'(t) = \frac{3}{8} + \frac{\pi}{32} \sin(2\pi t) > 0 \quad \forall t \in \mathbb{R}$$

$\therefore$ g is an increasing function in $[1,2] \Rightarrow$ minimum occurs when $t=0$ and $g(1)=1$

$$\text{hence } g(x) = \begin{cases} 1 & 0 \leq x < 1 \\ 1 & 1 < x \leq 2 \end{cases} = 1 \quad \forall x \in [0,2]$$

$\lim_{x \rightarrow 1^-} f \circ g(x) = f(1)$ and $\lim_{x \rightarrow 1^+} g \circ f(x) = 1$ also $f(1) > 1$

D06/3/184. Let $Z(x) = \frac{d}{dx} f(x)^{g(x)}$ and $Y(x) = \frac{d}{dx} g(x)^{f(x)}$ then $Z(x)$ and $Y(x)$ vanish simultaneously at :

- (A) $x = -\frac{1}{3}$ (B) $x = 0$ (C) $x = 1$ (D) No real value of x

माना $Z(x) = \frac{d}{dx} f(x)^{g(x)}$ तथा $Y(x) = \frac{d}{dx} g(x)^{f(x)}$ हैं तब $Z(x)$ तथा $Y(x)$ एक साथ शून्य होते हैं :

- (A) $x = -\frac{1}{3}$ (B) $x = 0$ (C) $x = 1$ (D) x का कोई वास्तविक मान नहीं

Ans. D

Sol. Consider $f(x)$ in $[0,1]$

$$f'(t) = 3t^2 - 2t + 1 > 0 \quad \forall t \in (0,1) \Rightarrow f \text{ is } \uparrow \text{ in } (0,1)$$

maximum occurs at $t=x$.

$$\text{and hence } f(x) \text{ is } \begin{cases} x^3 - x^2 + x + 1 & 0 \leq x \leq 1 \\ 3 - x & 1 < x \leq 2 \end{cases}$$

again consider $g(x)$ in $[0,1)$

$$g(t) = \frac{3}{8}t^4 + \frac{1}{2}t^3 - \frac{3}{2}t^2 + 1$$

$$g'(t) = \frac{3}{2}t^3 + \frac{3}{2}t^2 - 3t = \frac{3}{2}t(t^2 + t - 2) = \frac{3}{2}t(t-1)(t+2)$$

$g(t)$ = decreases in $[0,1) \Rightarrow$ maximum occurs when $t=0$ and $g(0)=1$

again consider $g(x)$ function in $[1,2]$

$$g'(t) = \frac{3}{8} + \frac{\pi}{32} \sin(2\pi t) > 0 \quad \forall t \in \mathbb{R}$$

$\therefore$ g is an increasing function in $[1, 2]$ $\Rightarrow$ minimum occurs when $t = 0$ and $g(1) = 1$

$$\text{hence } g(x) = \begin{cases} 1 & 0 \leq x < 1 \\ 1 & 1 < x \leq 2 \end{cases} = 1 \quad \forall x \in [0, 2]$$

$$Z(x) = \frac{d}{dx} f(x)^{g(x)} = \frac{d}{dx} f(x)^1 \quad \forall x \in [0, 1) \cup (1, 2]$$

$$= f'(x) \quad \forall x \in [0, 1) \cup (1, 2]$$

$$\text{and } Y(x) = \frac{d}{dx} g(x)^{f(x)} = \frac{d}{dx} 1^{f(x)} = 0 \quad \forall x \in [0, 1) \cup (1, 2]$$

Hence the functions $Y(x)$ and $Z(x)$ can vanish simultaneously at $f'(x) = 0$ which is not possible for any real x .

Paragraph for Question D06/3/185 to D06/3/186

Let f be a differentiable function satisfying

$$\lim_{y \rightarrow 0} \left(\frac{f(x+y)}{f(x)} \right)^{\frac{1}{y}} = e^{f(-x) \tan x}$$

Also right hand derivative at $x = 0$ exists for h , satisfying

$$g(x-y) = g(x)g(y) + h(x)h(y) \quad \forall x, y \in \mathbb{R}$$

and

$$h(x-y) = h(x)g(y) - g(x)h(y) \quad \forall x, y \in \mathbb{R}$$

Now, based on above information, answer the following questions

माना f एक अवकलनीय फलन $\lim_{y \rightarrow 0} \left(\frac{f(x+y)}{f(x)} \right)^{\frac{1}{y}} = e^{f(-x) \tan x}$ को संतुष्ट करता है तथा $x = 0$ पर h का दायाँ अवकलज भी

विद्यमान है तथा $g(x-y) = g(x)g(y) + h(x)h(y) \quad \forall x, y \in \mathbb{R}$ और $h(x-y) = h(x)g(y) - g(x)h(y) \quad \forall x, y \in \mathbb{R}$ को संतुष्ट करते हैं।

अब दी गई सूचनाओं के आधार पर निम्न प्रश्नों के उत्तर दीजिए

D06/3/185. The value of $f'(0)$ is equal to :

$f'(0)$ का मान बराबर होगा :

(A) 1 (B) 0 (C) -1 (D) 2

Ans. B

Sol. $\lim_{y \rightarrow 0} \frac{f(x+y) - f(x)}{f(x) \times y} = e^{f(-x) \tan x}$

$$\frac{f'(x)}{f(x)} = f(-x) \tan x$$

$$f'(x) = f(x) f(-x) \tan x$$

$$\text{put } x = -x$$

$$f'(-x) = -f(x) f(-x) \tan x$$

$$f'(x) + f'(-x) = 0$$

$$2f'(0) = 0$$

$$f'(0) = 0$$

D06/3/186. The value of $g'(0)$ is equal to :

$g'(0)$ का मान होगा :

(A) 0 (B) 1 (C) -1 (D) 2

Ans. A

Sol. $\because h(0) = 0$ & $g(0) = 1$

$$g(x-y) = g(x)g(y) + h(x)h(y)$$

$$g(0) = g^2(x) + h^2(x)$$

$$0 = 2g(x)g'(x) + 2h(x)h'(x)$$

$$\text{put } x = 0$$

$$g'(x) = 0$$

Paragraph for Question D08/3/187 to D08/3/188

Let $g(x)$ be a non-constant twice differentiable function defined on $\mathbb{R}$ (the set of all real numbers) such that $y = g(x)$ is symmetric about the line $x = 2$ and $g(-2) = g'\left(\frac{1}{2}\right) = g'(1) = 0$.

$y = g(x)$ is symmetric about the line $x = 2$ and $g(-2) = g'\left(\frac{1}{2}\right) = g'(1) = 0$.

यदि $g(x)$, $\mathbb{R}$ पर परिभाषित दो बार अवकलनीय फलन, जो अचर न हो इस प्रकार है, कि $y = g(x)$ रेखा $x = 2$ के सापेक्ष

सममित है तथा $g(-2) = g'\left(\frac{1}{2}\right) = g'(1) = 0$.

D08/3/187. The value of $g(6)$ equals :

$g(6)$ का मान होगा :

(A) $\frac{1}{2}$ (B) 1 (C) 2 (D) 0

Ans. D

Sol. We have $g(2-x) = g(2+x) \forall x \in \mathbb{R}$ (i)

Putting $x=4$, we get

$$g(-2) = g(6) = 0 \text{ (Given)}$$

D08/3/188. The minimum number of roots of the equation $g''(x) = 0$ in the interval $(0, 4)$ equals :

समीकरण $g''(x) = 0$ के अंतराल $(0, 4)$ में न्यूनतम मूलों की संख्या होगी :

- (A) 4 (B) 6 (C) 8 (D) 10

Ans. A

Sol. On differentiating (i) with respect to x , we get

$$-g'(2-x) = g'(2+x)$$

Now, putting $x = 0, 1, \frac{3}{2}$, we get

$$g'\left(\frac{1}{2}\right) = 0, g'(1) = g'(2) = g'(3) = g'\left(\frac{7}{2}\right)$$

So, clearly by using Rolle's theorem, for function $y = g'(x)$ in intervals $\left[\frac{1}{2}, 1\right]; [1, 2]; [2, 3]; \left[3, \frac{7}{2}\right]$

We conclude that the minimum number of roots of the equation $g''(x) = 0$ in $(0, 4)$ are 4.

Paragraph for Question D08/3/189 to D08/3/190

If $f(x) = \text{mid} \{g(x), h(x), p(x)\}$ means the function which will be second in order when values of the three

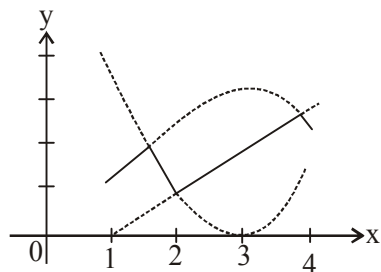
functions at a particular value of x are arranged, then for $f(x) = \text{mid} \left\{ x-1, (x-3)^2, 3 - \frac{(x-3)^2}{2} \right\}, x \in [1, 4]$.

D08/3/189. The numerical value of difference between the LHD and RHD of $f(x)$ at $x=2$ will be :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. D

Sol.



$$f(x) = x - 1 \quad x > 2$$

$$= (x-3)^2 \quad x < 2$$

$$\text{LHD} = -2 \text{ and } \text{RHD} = 1$$

D08/3/190. The greatest value of $f(x)$ in $[1, 4]$ will be :

- (A) $1+\sqrt{3}$ (B) $2+\sqrt{3}$ (C) $3+\sqrt{3}$ (D) $1+2\sqrt{3}$

Ans. A

Sol.
$$x-1 = 3 - \frac{(x-3)^2}{2} = \frac{-x^2 + 6x - 3}{2}$$

$$\Rightarrow x^2 - 4x + 1 = 0$$

$$\Rightarrow x = 2 + \sqrt{3} \text{ and greatest value of } f(x) = 1 + \sqrt{3}$$

Paragraph for Question D08/3/191 to D08/3/192

Let $f(x)$ is a cubic polynomial with leading coefficient unity such that $f(a) = b$ & $f'(a) = f''(a) = 0$. Let $\phi(x) = f(x) - f(a) + (a-x)f'(x) + 3(x-a)^2$ for which conclusion of Rolle's theorem in $[a, b]$ holds at $x = 2 \{a < 2 < b\}$.

माना घन बहुपद $f(x)$ जिसका मुख्य गुणांक इकाई है इस प्रकार है कि $f(a) = b$ तथा $f'(a) = f''(a) = 0$ है। माना $\phi(x) = f(x) - f(a) + (a-x)f'(x) + 3(x-a)^2$ जिसके लिये अन्तराल $[a, b]$ में रोले प्रमेय $x = 2$ पर सत्यापित होती है।

$$\{a < 2 < b\}$$

D08/3/191. $f''(2)$ is equal to :

$f''(2)$ का मान होगा :

- (A) 2 (B) 3 (C) 4 (D) 6

Ans. D

Sol.
$$f(x) = (x-a)^3 + b$$

$\therefore$ Rolle's theorem is applicable to $\phi(x)$ at $x=2$

$\therefore \phi(a) = \phi(b) \text{ \& } \phi'(2) = 0$

$$\Rightarrow 0 = f(b) - f(a) + (a-b)f'(b) + 3(b-a)^2$$

$$\text{\& } (a-2)f''(2) + 6(2-a) = 0$$

$$\Rightarrow f'(b) = \frac{f(b)-f(a)}{b-a} + 3(b-a) \text{ \& } f''(2) = 6$$

D08/3/192. Maximum value of $\phi(x)$ in $[a, b]$ is :

अन्तराल $[a, b]$ में $\phi(x)$ का अधिकतम मान होगा :

- (A) 0 (B) 1 (C) 2 (D) $a+b$

Ans. B

Sol. $f''(2) = 6 \Rightarrow 6(2 - a) = 6 \Rightarrow a = 1$ &

$$f'(b) = \frac{(b-a)^3}{b-a} + 3(b-a) \{ \because f(x) = (x-a)^3 + b \}$$

$$\Rightarrow 3(b-a)^2 = (b-a)^2 + 3(b-a)$$

$$2(b-a)^2 = 3(b-a)$$

$$b-a = \frac{3}{2}$$

$$\Rightarrow b = \frac{5}{2}$$

Let $f'(\sin x) < 0$ and $f''(\sin x) > 0 \forall x \in \left(0, \frac{\pi}{2}\right)$ and $g(x) = f(\sin x) + f(\cos x)$.

Paragraph for Question D08/3/193 to D08/3/194

Let $f(x)$ be twice differentiable function everywhere such that $f(0) = 0$, $f(1) = 9$ and $f(x) + f''(x) > 2f'(x) \forall x \in [0, 1]$.

माना $f(x)$ दो बार सर्वाश अवकलनीय फलन इस प्रकार है कि $f(0) = 0$, $f(1) = 9$ तथा $f(x) + f''(x) > 2f'(x) \forall x \in [0, 1]$

D08/3/193. Which of the following statement is/are true about $f(x)$?

निम्न में से कौनसा/कौनसे फलन $f(x)$ के सापेक्ष सत्य है ?

(A) $f(x) \geq 9xe^{x-1} \forall x \in [0, 1]$

(B) $f(x) \leq 9xe^{x-1} \forall x \in [0, 1]$

(C) $f(x) \geq 27xe^{x+1} \forall x \in [0, 1]$

(D) इनमें से कोई नहीं

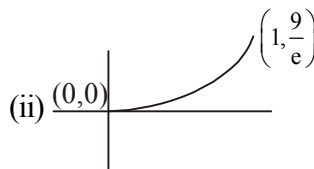
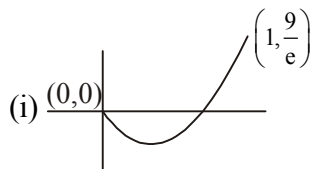
Ans. B

Sol. $f(x) + f''(x) - 2f'(x) > 0 \forall x \in [0, 1]$

$$\Rightarrow e^{-x}(f(x) + f''(x) - 2f'(x)) > 0 \forall x \in [0, 1] \Rightarrow [e^{-x}f(x)]'' > 0 \forall x \in [0, 1]. \text{ Let } h(x) = e^{-x}f(x)$$

$$\Rightarrow h(0) = 0, h(1) = \frac{9}{e}, h''(x) > 0 \Rightarrow h(x) \text{ is concave upward/h(x) की अवकलता ऊपर की ओर है}$$

Two possible graphs of $h(x)/h(x)$ के दो संभावित आरेख



Now equation of line joining $(0, 0)$ and $\left(1, \frac{9}{e}\right)$ is $y = \frac{9x}{e}$

अब $(0, 0)$ और $\left(1, \frac{9}{e}\right)$ रेखा का समीकरण $y = \frac{9x}{e}$ है।

$$\Rightarrow h(x) \leq \frac{9x}{e} \quad \forall x \in [0, 1] \quad \Rightarrow e^{-x} f(x) < \frac{9x}{e} \quad \forall x \in (0, 1)$$

$$\Rightarrow f(x) < 9xe^{x-1} \quad \forall x \in (0, 1)$$

D08/3/194. Let $h(x) = \frac{f(x)}{e^x}$, then which of the following statement is true ?

माना $h(x) = \frac{f(x)}{e^x}$, तब निम्न में से कौनसा सही है ?

- (A) $h\left(\frac{4}{9}\right) \leq \frac{4}{e}$ (B) $h\left(\frac{4}{9}\right) > \frac{4}{e}$ (C) $h\left(\frac{4}{9}\right) > 2$ (D) इनमें से कोई नहीं

Ans. A

Sol. The point which divides $(0,0)$ and $\left(1, \frac{9}{e}\right)$ in the ratio 4 : 5 internally is $\left(\frac{4}{9}, \frac{4}{e}\right)$

$(0,0)$ और $\left(1, \frac{9}{e}\right)$ को बिन्दु 4 : 5 में आंतरिक विभाजित करने वाला बिन्दु $\left(\frac{4}{9}, \frac{4}{e}\right)$ है।

$$\Rightarrow h\left(\frac{4}{9}\right) \leq \frac{4}{e}$$

Paragraph for Question D08/3/195 to D08/3/196

$f(x)$, $g(x)$ and $h(x)$ are all continuous and differentiable increasing functions in $[a, b]$. Also $a < c < b$ & $f(a) = g(a) = h(a)$. Further the point of intersection of tangent at $x = c$ with the chord joining $x = a$ & $x = b$ is on the left of c in $y = f(x)$ and on the right of c in $y = h(x)$. In case of $g(x)$, tangent is parallel to the chord. Given $f(x)$, $g(x)$, $h(x)$ are concave up.

$f(x)$, $g(x)$ और $h(x)$ सभी $[a, b]$ में सतत और अवकलनीय वर्धमान फलन है तथा $a < c < b$ और $f(a) = g(a) = h(a)$. पुनः $x = a$ और $x = b$ को मिलाने वाली जीवा के साथ $x = c$ पर स्पर्श रेखा का प्रतिच्छेद बिन्दु, वक्र $y = f(x)$ में c के बांयी ओर तथा $y = h(x)$ में c के दांयी ओर है। $g(x)$ की स्थिति में स्पर्शरेखा जीवा के समान्तर है। दिया गया है कि $f(x)$, $g(x)$, $h(x)$ ऊपर की ओर अवतल है।

D08/3/195. If $f'(x) > g'(x) > h'(x)$, then :

यदि $f'(x) > g'(x) > h'(x)$, तब :

- (A) $f(b) < g(b) < h(b)$ (B) $f(b) > g(b) > h(b)$ (C) $f(b) = g(b) = h(b)$ (D) $f(b) \leq g(b) \leq h(b)$

Ans. B

Sol.

D08/3/196. If $f(b) = g(b) = h(b)$, then :

(A) $f'(c) = g'(c) = h'(c)$

(B) $f'(c) > g'(c) > h'(c)$

(C) $f'(c) < g'(c) < h'(c)$

(D) None of these

यदि $f(b) = g(b) = h(b)$, तब :

(A) $f'(c) = g'(c) = h'(c)$

(B) $f'(c) > g'(c) > h'(c)$

(C) $f'(c) < g'(c) < h'(c)$

(D) इनमें से कोई नहीं

Ans. C

Sol. Applying LMVT से

$$\frac{f(b) - f(a)}{b - a} > f'(c); \quad \frac{g(b) - g(a)}{b - a} = g'(c) \text{ and और } \frac{h(b) - h(a)}{b - a} < h'(c)$$

$$\text{As चूँकि } f'(x) > g'(x) > h'(x) \forall x \in R \quad \Rightarrow \quad f(b) > g(b) > h(b)$$

Paragraph for Question D08/3/197 to D08/3/198

A function $f: R \rightarrow R$ satisfies the following properties :

$$f'(x) > 0 \forall x \in (-\infty, -2) \cup (3, \infty), \quad f'(x) \leq 0 \forall x \in (-2, 3), \quad \lim_{x \rightarrow \infty} f(x) = 0$$

$$f \text{ is continuous } \forall x \in R - P\{3\}, \quad f \text{ is differentiable } \forall x \in R - \{-2, 3\}, \quad \lim_{x \rightarrow 3} f(x) \rightarrow -\infty,$$

$$f''(x) > 0 \forall x \in (-\infty, 0) - \{-2\}, \quad f''(x) < 0 \forall x \in (0, \infty) - \{3\}, \quad \lim_{x \rightarrow -\infty} f(x) = 3,$$

and $f(0) = 0$. Now, answer the following questions :

D08/3/197. Let λ be the maximum number of solutions of $f(x) = |x|$. If

$$f'(0) > -3 \text{ and } f(-2) > 6,$$

then number of solutions of $f(x) + 3x = 0$ is equal to :

(A) $\lambda + 1$

(B) $\lambda - 1$

(C) $2\lambda - 1$

(D) $3\lambda - 4$

Ans.

Sol. A

D08/3/198. The function $y = f(-|x|)$ is :

(A) differentiable $\forall x \in R$, if $f'(0) = 0$

(B) continuous but NOT differentiable at two points, if $f'(0) = 0$

(C) continuous but NOT differentiable at one point, if $f'(0) = 0$

(D) dis-continuous at two points if $f'(0)=0$

Ans. B
Sol.

Paragraph for Question D08/3/199 to D08/3/200

Let $|X|$ denotes the cardinal number of set X ,

$$A = \{27\lambda \in \mathbb{R} \mid |B| = 3\}, B = \{x \in \mathbb{R} \mid x^3 + x^2 - x + \lambda = 0\} \text{ and } f(x) = 2 - \sqrt{4 - (x-1)^2}.$$

Now, answer the following questions :

D08/3/199. Which of the following options lie in set A ?

- (A) 5 (B) -16 (C) -27 (D) 4

Ans. BD
Sol.

D08/3/200. Which of the following options is/are correct ?

- (A) f is monotonic in interval $(-1, 1)$ (B) f is monotonic in interval $(1, 3)$
(C) $y = 2x - 2\sqrt{5}$ is tangent to f (D) $y = -3x + 5 + 2\sqrt{10}$ is tangent to f

Ans. ABC
Sol.

Paragraph for Question D07/3/201 to D07/3/203

$$\text{Let } f(x) = \begin{cases} 1-x & ; 0 \leq x \leq 1 \\ 0 & ; 1 < x \leq 2 \\ (2-x)^2 & ; 2 < x \leq 3 \end{cases} \text{ and } g(x) = \int_0^x f(t) dt.$$

Let the tangent to the curve $y = g(x)$ at point P whose abscissa is $\frac{5}{2}$ cuts x -axis in point Q .

Let the perpendicular from point Q on x -axis meets the curve $y = g(x)$ in point R .

D07/3/201. $g(1) = :$

$g(1) = :$

- (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2

Ans. B

Sol. $g(1) = \int_0^1 f(t) dt$

$$= \int_0^1 (1-t) dt = t - \frac{t^2}{2} \Big|_0^1 = \frac{1}{2}$$

D07/3/202. Equation of tangent to the curve $y = g(x)$ at P is :

- (A) $3y = 12x + 1$ (B) $3y = 12x - 1$ (C) $12y = 3x - 1$ (D) $12y = 3x + 1$

Ans. C

Sol. For $x \in (2, 3]$

$$g(x) = \int_0^1 f(t) dt + \int_1^2 f(t) dt + \int_2^x f(t) dt$$

$$g(x) = \frac{1}{2} + \frac{(x-2)^3}{3}$$

$$\text{at } x = \frac{5}{2}, g\left(\frac{5}{2}\right) = \frac{13}{24}$$

$$g'(x) = f(x)$$

$$g'\left(\frac{5}{2}\right) = \frac{1}{4}$$

$$y - \frac{13}{24} = \frac{1}{4} \left(x - \frac{5}{2} \right)$$

$$12y = 3x - 1$$

D07/3/203. If ' θ ' be the angle between tangents to the curve $y = g(x)$ at point P and R; then $\tan\theta$ equals to :

- (A) $\frac{5}{6}$ (B) $\frac{5}{14}$ (C) $\frac{5}{7}$ (D) $\frac{5}{12}$

Ans. B

Sol. Slope of tangent at P = $\frac{1}{4}$

Slope of tangent at R = $\frac{2}{3}$

$$\tan\theta = \frac{5}{14}$$